

Senior Project Defense

Xecure Password Manager

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Problem Statement

The management of passwords is one of the most important aspects of Cybersecurity and it is imperative that users implement and practice safe password management practices. In order to ensure safe password management practices users must create strong and unique passwords for every one of their accounts. Unfortunately, most users have far too many online accounts to remember a unique password for each account. A survey conducted by Nate Lord (2020) indicates that the average user has over 90 different online accounts. There is a limit to how many passwords a user can remember, and this survey presents a problem that can lead to users reusing the same passwords. A Google Online Security Survey (2019) found that 52 percent of users reuse the same password for multiple accounts. The practice of reusing passwords leaves users vulnerable to intrusion attacks, which can result in data loss, monetary loss and more. A report conducted by IBM (2020) found that the average cost of a data breach is \$3.68 million, while a data breach in the healthcare industry is estimated to cost around \$7.13 million. Furthermore, IBM found that data breaches in the United States are among the costliest averaging around \$8.64 million.

By using a password manager users can easily manage unique and strong passwords for their accounts while eliminating all the difficulties that are associated with it. With a password manager users can store all their passwords in a secured vault that can be accessed with a “master” password. This eliminates the need to remember multiple passwords and greatly reduces the chances of having multiple accounts compromised. The management of passwords is one of the most important and overlooked factors of information security. In order to eliminate the difficulties that come with remembering passwords it is recommended that users utilize a password management application.

Project Description

The project is a password manager application for Windows computers used to securely store and manage log in credentials. It uses a “master” password to provide access into the user’s account with all the records and credentials. The application also has connectivity to a MySQL database that is used to securely store the user’s information. In order to protect the information in the database the application uses the well-established AES 256 encryption standard and the SHA 256 hashing algorithm. Additionally, the users have the option to generate unique and complex passwords for every record entered. Users will also have the option of entering their own passwords and the application will guide them every step of the way. The application can also copy a record’s password into the computer’s clipboard. This password will then be wiped from the clipboard after fifteen seconds or if the user closes the application. Lastly, the program has features that allows users to reset their “master” password and email if they forget it or wish to change it.

The application is called Xecure Password Manager and it was inspired by applications like KeePass, and LastPass and it contains all the major features that these application have. All of the previously mentioned features are the standard in the password manager market, and they will be included in the Xecure Password Manager application at no cost to the users. In addition to being a free application, Xecure Password Manager will not contain any ads, there are no password or account limits, and it will not store or use user data for any other purpose. This is what separates Xecure Password Manager from the competition.

Personal Motivation

The practice of securing networks, systems, and other digital infrastructures are an essential aspect of information systems and cybersecurity. As a Cybersecurity major, I strive to broaden these previously mentioned technical and functional skills. Through the development of this application, I was hoping to get a better understanding of the various hashing and encryption practices that are common in the field of cybersecurity. Another skill I wanted to learn was database management. I have made databases in the past, but I never got the opportunity to use or manage my own databases. I also wanted to strengthen my problem solving and debugging skills that are so vital as a programmer. Lastly, I wanted to learn how create and prevent SQL injections.

This project required me to learn and understand the various methods that attackers use to exploit passwords and databases. Through this project I was able to expand my computer analysis skills, and it gave me additional resources that can be used to thwart malicious attacks. I believe that after completing this project I have become a better cybersecurity professional.

Research & Background

In order to create this application, extensive research was done into the password manager industry. Xecure was inspired by applications like KeePass and LastPass and it is meant to be an alternative to these applications. In order to do this, it must offer the same features as the most popular password manager applications in the industry. After conducting research, the current and more popular password managers in the market were found to be:

- KeePass
- LastPass
- 1Password

- DashLane
- NordPass

All of these applications are very similar and contain the following features that I see as an industry standard:

- Encryption of passwords
- Hashing of credentials
- Save password to clipboard
- Wipe passwords from clipboard
- Generate strong and unique passwords
- Support multiple users
- Store multiple records
- Email authentication

The previously mentioned features will all be part of the Xecure Password Manager application and will be offered to the users at no charge. Additionally, other features will be added to separate the application from the competition.

Proposed Implementation Languages

The password manager application was developed on a Windows 10 computer by using the latest versions of Python 3 (version 3.9.7) and MySQL Server (version 8).

Libraries, packages, development kits

The password manager application uses the following libraries:

- hashlib – Used for hashing.
- cryptography.fernet – Used to encrypt.
- msvcrt – Used to prevent the command screen from continuing until a key is pressed.
- mysql.connector – Used to connect to the database.
- os – Used to change the title of the command line screen.

- sys – Used alongside getpass to hide passwords in the command line screen.
 - getpass – Used to hide password in the command line screen.
 - random – Used to generate random passwords and codes.
 - smtplib – Used for emailing capabilities.
 - socket – Used alongside smtplib to send emails, and email error handling.
 - time – Used alongside pyperclip to determine when to wipe clipboard.
 - pyperclip – Used to save record's password into the computer's clipboard.
 - threading – Used to wipe the clipboard while the user continues to use the application.
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Additional Software/Equipment Needed

A Windows operating system is required, as well as Python (version 3.9.7) and a MySQL (version 8) database. See below for instructions on how to download and setup both Python and MySQL.

MySQL Installation Guide:

These steps will guide you through the installation and setup of the MySQL Community Server for the required database of this application.

1. Visit <https://dev.mysql.com/downloads/mysql/>
2. Look for the “MySQL (MSI) Installer for Windows” and click on the “Go to Download Page” button (You can also use this link:
<https://dev.mysql.com/downloads/windows/installer/8.0.html>)
3. There will be two Windows MSI Installer options here. Please find the “(mysql-installer-community-8.0.27.0.msi)” option and press the download button.
4. You will be redirected to a screen that asks you to login or sign up, but this is not needed. Scroll down to the bottom of the page and click on the “No thanks, just start my download.” link.

5. A download prompt will be displayed. Click on the “Save File” button. This will start the download of the file.
6. Wait for the download to finish.
7. Once it has downloaded, look for the “mysql-installer-community-8.0.26.0” (version number might change) file and run it. The file is typically located in your downloads folder.
8. After double clicking the file, it will begin preparing the installation and it will then ask you for permission to install the application. Click the “yes” button.
9. It may ask you “Do you want to allow this app to make changes to your pc?”. Click on the “yes” button. (Skip this step if you did not see this alert)
10. The MySQL installation windows will now be displayed, and it will ask you to read over the License Agreement. After doing so, press the “I accept the license terms” check box and press the “Next” button.
11. On the “Choosing a Setup Type” screen select the “Developer Default” option and press next.
12. A screen may alert you that some product requirements have not been satisfied and it will ask you if you wish to continue. Select yes. (Skip this step if you did not see this alert)
13. On this installation screen you can see all of the products that will be installed. When you are ready select the “Execute” button. This will begin installing all of the products.
14. Wait for the Installation to finish.

15. After the download has finished you will see a green checkmark on all of the products downloaded. If everything downloaded successfully press the “Next” button.
16. On the “Product Configuration” screen you will be able to see the configuration for all of your downloaded products. Once you have reviewed this information press the “Next” button.
17. On the “Type and Networking” screen select the “Standalone MySQL Server/Classic MySQL Replication” option and press the “Next” button.
18. On this screen leave the “Config Type” as “Development Machine”. Make sure that the “TCP/IP” checkbox and “Open Firewall port for network access” checkbox are selected. Lastly, the “Port Number” is 3306. After confirming these options press the “Next” button.
19. On the Accounts and Roles screen enter a strong password on the “MySQL Root Password” fields (**make sure you remember this password**). Then press the “Add User” button to create a user account.
20. A screen will appear and ask you for username information. Enter a username for your account (**make sure you remember this username**). Then enter a strong password for your account (**make sure you remember this password**). Once this is done press the “Ok” button. After creating your user account press on the “Next” button.
21. On the “Windows Service” screen leave all settings as default and press on the “Next” button.

22. On the “Plugins and Extensions” screen leave all settings as default and press on the “Next” button.
23. On the “Apply Configuration” screen you will see an overview of the configurations. Once you are ready press the “Execute” button and the listed changes will be applied. Once this is done press the “Finish” button.
24. On the “Product Configuration” Screen you will see that the MySQL Server is completed. Press the “Next” button to continue.
25. On the “MySQL Router Configuration” screen you will leave all settings as default and press on the “Next” button.
26. On the “Product Configuration” screen you will review the configuration and press on the “Next” button.
27. On the “Connect To Server” screen, select the “MySQL Server 8.0.27.0 (version may be different for you) text. Then enter press the “Check” button to test the connection. Once the connection is established and you see a “Connection Successful” message press the “Next” button.
28. On the apply configurations, press the “Execute” button. Once all the configurations are finished press the “Finish” button.
29. On the “Product Configuration” screen you will review the configuration and press on the “Next” button.
30. You will see the installation complete screen. Press the “Finish” button.
31. You have successfully installed the MySQL database.

To use with MySQL Workbench:

1. Go to start and search for “workbench”
2. Open the MySQL Workbench
3. Once the application opens, double click the “Local instance” below the “MySQL Connections”.
4. A screen will appear, and you will be prompted to enter the password for the database. After entering the password, press the “OK” button.
5. This is your database serve.

Requirements

Requirement ID Number: 1 Requirement Type: 1, 2 (Functional, Look and Feel)

Description: The application will have a command-line-based user interface.

Rationale: An interface will be needed so that users can interact with the application in an efficient and simple manner. A broader audience will be able to use the application if a command-line-based user interface is implemented.

Fit Criterion: Users must be able to use all features and accomplish all tasks by using the command-line-based interface.

Priority: Very High

Dependencies: None

Requirement ID Number: 2 Requirement Type: 1, 6 (Functional, Security)

Description: A login section must be implemented.

Rationale: The login section is needed to authenticate users and grants them access to their data.

Fit Criterion: When presented with valid credentials the application must authenticate users and grant them access to their data. If presented with invalid credentials, the application will display an error message and will not grant them access to the data. Users will not be able to get past the login screen without valid credentials.

Priority: Very High

Dependencies: None

Requirement ID Number: 3

Requirement Type: 1, 6 (Functional, Security)

Description: The application must have connectivity to a MySQL database.

Rationale: The database provides storage and security features.

Fit Criterion: The application must be able to save data to the database. The data will be inaccessible if there is no database connectivity. Additionally, it will not allow the user to create a record if the database cannot be reached.

Priority: Very High

Dependencies: None

Requirement ID Number: 4

Requirement Type: 6 (Security)

Description: The application must store user records in the MySQL database.

Rationale: The database provides storage and security features. It also provides features against data redundancy and reduces inconsistencies. All these features are important for managing log-in credentials.

Fit Criterion: The application must be able to save data to the database. This data must then be accessible to authorized users only. An error message will alert the user if the data was not saved to the database.

Priority: Very High

Dependencies: The application must have connectivity to the database for this requirement to be fulfilled.

Requirement ID Number: 5 **Requirement Type:** 1, 6 (Functional, Security)

Description: A “Register” section must be implemented.

Rationale: Users need to be able to create an account to use the application. Therefore, a “Register” section must be implemented.

Fit Criterion: The “Register” section must allow users to create a master username and password. If correctly done the username and password will be stored in the database. These credentials can then be used to log in to the application. Users cannot use the application without creating an account.

Priority: Very High

Dependencies: A “Login” section and the database must be implemented before this requirement can be fulfilled.

Requirement ID Number: 6 **Requirement Type:** 6 (Security)

Description: Application will lock-out an account after six failed log-in attempts and the password will be reset.

Rationale: Unauthorized users may attempt to guess the login credentials of a user. To prevent these brute force attacks a lock-out feature will be implemented.

Fit Criterion: After six failed log-in attempts the application will place a lock on the account and the user will not be able to log in. The lock-out feature will cause the application to force close.

Priority: Very High

Dependencies: The “Login” section and the “Register” section must be implemented before this requirement can be fulfilled.

Requirement ID Number: 7

Requirement Type: 6 (Security)

Description: All data stored and used by the application must only be accessible to authorized users.

Rationale: The data stored and used by the application must only be visible to authorized users. To prevent a data breach, registered users will only have access to their data. Users that are not registered will not be able to use the application and will not have access to any data.

Fit Criterion: Registered users must only be able to see their data, while unregistered users will not be able to use the application.

Priority: Very High

Dependencies: The “Login”, “Register”, and database sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 8

Requirement Type: 6 (Security)

Description: A SHA256 hash will be used as a log-in authentication method.

Rationale: In order to improve security, the username and password along with a hashing algorithm will be used to authenticate the user.

Fit Criterion: The SHA256 will be used to hash the username and password provided. If the username and password provided match the login credential stored in the database, the user will be authenticated, and access will be granted. If the hash does not match the login credential stored in the database access to the application will be denied.

Priority: Very High

Dependencies: The “Login”, “Register”, and database sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 9 **Requirement Type:** 3, 6 (Usability, Security)

Description: The application must be able to generate random passwords.

Rationale: Users may need help creating secure passwords. The application will provide them with the option of generating a strong random password for their desired account.

Fit Criterion: A random password must be generated and displayed to the user after opting for this feature. An error will alert the user if the random password was not generated.

Priority: Medium

Dependencies: None

Requirement ID Number: 10 **Requirement Type:** 3, 6 (Usability, Security)

Description: The application will allow users to create their own password.

Rationale: Users may not like having randomly generated passwords. They may want to use their own passwords.

Fit Criterion: Users must be able to enter their own passwords. A confirmation message will alert the user that their password was accepted and saved.

Priority: High

Dependencies: The “Login”, “Register”, and database sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 11 **Requirement Type:** 3, 6 (Usability, Security)

Description: The rules for the record passwords cannot be strict

Rationale: The application cannot dictate what users have chosen in the past for their passwords. It must accept all kinds of passwords no matter how weak or strong.

Fit Criterion: Any password less than thirty-five characters long will be accepted as a record password, but all other password combinations are valid. The user will be notified if the password was saved successfully.

Priority: High

Dependencies: The “Login”, and “Register” sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 12 **Requirement Type:** 3 (Usability)

Description: Simple controls must be implemented on the front end to send SQL queries to the database.

Rationale: Users do not want to create SQL queries for every task while using the application.

Fit Criterion: Queries will be created in the back end and the users must be able to utilize them on the front end. Users must be able to view, add, edit, and delete records on the database from the front end. A confirmation message will be displayed after a successful query.

Priority: High

Dependencies: The “Login”, and “Register” sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 13 **Requirement Type:** 3 (Usability)

Description: Data stored in the database will be encrypted by using AES 256 encryption.

Rationale: Encrypting the data prevents unauthorized users from using it. Encryption increases the security of the application and protects the data.

Fit Criterion: No plaintext must be visible in the database. If plaint text is found in the database this requirement has not been fulfilled. The information in the database must be decrypted before it can be used. If an encryption attempt fails an error message will be displayed to the screen.

Priority: Very High

Dependencies: The database must be implemented before this requirement can be fulfilled.

Requirement ID Number: 14 **Requirement Type:** 3, 6 (Usability, Security)

Description: Registered users must be able to retrieve their unencrypted data from the database.

Rationale: Users need to be able to see their saved unencrypted login credentials.

Fit Criterion: Users must be able to select a specific account to retrieve their unencrypted password. If the data retrieved is encrypted this requirement has not been fulfilled. If the data cannot be retrieved or if it cannot be unencrypted an error message will be displayed to the user.

Priority: High

Dependencies: The database and the encryption algorithm must be implemented before this requirement can be fulfilled.

Requirement ID Number: 15 Requirement Type: 3 (Usability)

Description: Passwords retrieved from the database must be copied to the user's clipboard.

Rationale: This is done so that the user can immediately use the password. After retrieving the password users can paste it into their desired field.

Fit Criterion: After retrieving the data the user must be able to use the "paste" command and the password will be "pasted" into their desired field. An error message will be displayed if the password was not placed in the clipboard. On the other hand, a confirmation message will be displayed if the password was placed into the user's clipboard.

Priority: Low

Dependencies: The "Login", "Register", database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 16 Requirement Type: 3 (Usability)

Description: The application must guide and direct users on how to accomplish all tasks.

Rationale: Users may not understand how to use the application. Clear and concise instructions will help the user navigate the application.

Fit Criterion: Prompts and alerts will guide the user on how to accomplish their task. The success of the directions will be based on task completion time. The average task completion time should be under five minutes.

Priority: Medium

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 17 Requirement Type: 3 (Usability)

Description: The users must be able to go back to the previous menu at any point of the process.

Rational: The users may decide they wants to cancel their current action.

Fit Criterion: A back option must be present in every part of the application. Once the option has been selected the user will be taken back to the previous screen.

Priority: Medium

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 18 Requirement Type: 6 (Security)

Description: The users must be able to log out of the application.

Rational: The users need to be able to log out of the application in a secure manner.

Fit Criterion: Once the option has been selected an alert message will prompt the user to confirm the action. The user will be logged out of the application if they confirm the action.

Priority: Very High

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 19 Requirement Type: 3 (Usability)

Requirement ID Number: 21 Requirement Type: 6 (Security)

Description: Input validation must be performed throughout every section of the application.

Rational: Users may enter invalid data accidentally or purposefully and thus every input must be properly validated and sanitized to prevent a security risk. All prompts will be validated and will include exceptions to handle all scenarios.

Fit Criterion: All invalid inputs will produce an error message that will be displayed to the screen.

Priority: Very High

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 22 **Requirement Type:** 3 (Usability)

Description: Error messages must provide guidance on how to complete the task.

Rational: Users may not understand why they can’t proceed and may need help fixing their mistakes.

Fit Criterion: The error messages must provide clear and concise instruction on how to complete the current task. The success of the directions will be based on task completion time. The average task completion time should be under five minutes.

Priority: Low

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 23 **Requirement Type:** 6 (Security)

Description: The application must have measures in place to prevent SQL injection attacks.

Rational: Unauthorized users may attempt to use SQL injections to retrieve information from the database.

Fit Criterion: Inserting SQL injections will result in an invalid input error.

Priority: Very High

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 24 **Requirement Type:** 6 (Security)

Description: The average user should be able to retrieve a password in under three minutes.

Rational: Users typically won’t use an application that takes too long to complete a task.

Fit Criterion: Users must be able to retrieve a password in under three minutes.

Priority: Medium

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 25 **Requirement Type:** 6 (Security)

Description: The application must have emailing capabilities in order to reset passwords.

Rational: The application may need to send emails to the users. To reset passwords, the application will require emailing capabilities.

Fit Criterion: Application must be able to send emails to the users. Once the email has been sent a confirmation message will be displayed to the screen.

Priority: Medium

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 26

Requirement Type: 6 (Security)

Description: A verification code must be sent to the users email in order to verify their account.

Rational: Users may forget their password or may need to change their email. Before doing so, users must verify their account by entering the code sent to their email.

Fit Criterion: Users must enter the correct code before they can change their password or email. If the entered code does not match the user will be given an error message. If the code matches, they will be able to change their password or email.

Priority: Very High

Dependencies: The “Login”, “Register”, database, emailing and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 27

Requirement Type: 6 (Security)

Description: Errors in the middle of data entry will display an error and no data will be saved to the database.

Rational: This requirement will prevent the application from saving invalid or incomplete data into the database. The error will inform the user what went wrong and how to fix the problem.

Fit Criterion: An error message will be displayed if something goes wrong in the middle of data input. The user will have to acknowledge the error before moving on.

Priority: High

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 27 Requirement Type: 6 (Security)

Description: Clipboard will be cleared after fifteen seconds.

Rational: The clipboard can be exploited to retrieve sensitive data. Thus, the clipboard must be wiped after a certain period of time.

Fit Criterion: After fifteen seconds the user will not be able to paste their password. Trying to paste a password after 15 seconds will result in an empty character.

Priority: Very High

Dependencies: The “Login”, “Register”, clipboard, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 28 Requirement Type: 6 (Security, Usability)

Description: Users must be able to retrieve their username

Rational: Users may forget their username.

Fit Criterion: Users should be able to retrieve their username by verifying their email account. After this verification an email will be sent to the user with their registered username.

Priority: Very High

Dependencies: The “Login”, “Register”, clipboard, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 28

Requirement Type: 6 (Usability)

Description: Users must be able to display only the name of their records.

Rational: Users may not want to display all the information of their records, and thus they should be able to retrieve only the record names. This name can later be used to retrieve the specific record's information.

Fit Criterion: Users should be able to see all their record names through the "Retrieve Record Names Only" screen. No record names will be seen if the database is empty.

Priority: Very High

Dependencies: The "Login", "Register", clipboard, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 29

Requirement Type: 6 (Usability)

Description: The application cannot be accessed if there are no registered users.

Rational: The application should not be accessible if no user has registered.

Fit Criterion: A "Database is empty! Please register to use the application." message will be displayed if no user has registered.

Priority: Very High

Dependencies: The "Login", "Register", database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 30

Requirement Type: 6 (Usability)

Description: Users should be able to delete all their records.

Rational: Users should have an option to delete all their records stored in the database. They should not have to do it one at a time.

Fit Criterion: All records must be deleted from the database and application if the user selects and confirms the “Delete all records” option.

Priority: Very High

Dependencies: The “Login”, “Register”, delete all, database, and encryption sections must be implemented before this requirement can be fulfilled.

Requirement ID Number: 31 **Requirement Type:** 6 (Security)

Description: All SQL statements will be composed of prepared statements

Rational: Using prepared statement protects the application from SQL injections.

Fit Criterion: Users should not get any information when attempting to perform SQL injections.

Priority: Very High

Dependencies: The “Login”, “Register”, database, and encryption sections must be implemented before this requirement can be fulfilled.

Additional Requirements:

Securing Database (Security)

In order to secure the database and the information within it, access into the database will be limited and will require a password to authenticate. Ideally only the root user will have control and access to the database. Other accounts will need well defined rules for database access, views access, and data storing procedures based on needs and the role of the individuals. To protect the information within the database, all the data will be

encrypted before being stored in the database. In order to prevent unauthorized connections into the database, access will be limited to local host. Additionally, in order to prevent SQL injections, all queries will be pre-defined within the application, and then the parameters will be passed on to the query after being sanitized. This will validate the user input and prevent users from creating their own queries.

Clipboard Vulnerability (Security)

Most Windows password manager applications utilize the clipboard to allow users to copy and paste a specified account password. This is a convenient feature as it prevents users from having to type long and complicated passwords, but it can also be exploited if the computer was infected with malware. When the password is copied into the clipboard this data is temporarily stored in the memory of the computer. The longer the password is kept in the clipboard the longer it will remain in memory. This creates a window of opportunity for an attacker to access the system and retrieve the password from the clipboard or the memory of the computer. To prevent this from happening password managers flush the clipboard after a certain amount of time has passed. A feature like this will be created and implemented in this password manager. The application will alert the user that the password will be wiped from the clipboard after fifteen seconds of retrieving the password. The user may retrieve the password again if they need it after the first fifteen seconds. In addition to clearing the clipboard after fifteen seconds, it will also be cleared when the user logs out, and lastly when the user exits the application. This feature will not completely prevent this vulnerability, but it will reduce the probability of it happening and it is the industry standard.

Project Implementation Description

The initial setup of the application consists of modifying some values within the code and creating an email for the application to use. First, user's need to edit the "database.py" file and add their database username on line four, and then the database password on line five. For more details, please see the example below:

Line 4: 'user': 'root',

Line 5: 'passwd': '12345',

Note: Replace "root" with your username and "12345" with your password.

The next step is to add an email for the database to use. It's recommended that a new email is created solely for the purpose of running this application. Once the email has been created, users need to modify line twenty-nine of the "passwordRecovery.py" file and add the email address and password of the new email that the application will use. For more details, please see the example below:

Line 29: s.login("xecure@gmail.com", "123")

Note: Replace "xecure@gmail" with your email and "123" with your password.

After making these changes the application will be ready for use. To run the application users, must run the "main.py" file. This will create the MYSQL database along with all the files necessary to start and run the Xecure Password Manager Application.

Once the application is started users will be greeted with the "Main Menu". From this menu users can log into the application by pressing "1", create an account by pressing "2", get help with their login information by pressing "3", or close the application by pressing "4" (see

figure 1 for reference). If the user has not registered, they will not be able to access the “Login” or “Forgot Login” sections. In this case a message will be displayed letting the user know that the database is empty and that they need to register to use the application. If the user already has an account, the “Login” screen will prompt the user to enter their username and their password (see figure 2 for reference). However, if the user wishes to go back to the previous screen, they can enter the “0” command, and it will take them back to the previous screen. This command can be used all throughout the application. In order to authenticate users a SHA256 hash is created from their username and password. If this hash matches what is on the database, they are successfully logged in to the application. If the user fails to provide the proper username and password after six login attempts, the application closes itself. A warning is given to the user after five failed login attempts.



Figure 1: Main Menu Screen



Figure 2: Login Screen

In order to use the application, users need to create an account. This can be done through the “Register” menu. This section will ask users to enter an email address, a username, and a password. The email address and the username must be unique, and the password must be longer than eight characters, it must contain both upper and lowercase letter, and it must contain numbers and symbols. Once a valid email, username, and password are entered the user will be greeted with a “Thank you for registering message” (see figure 3 and 4 for reference). They can then use this information to log into the application.

If a user is having difficulties login into the application, they can access the “Login Help” menu where they will be given the option to retrieve their username or reset their password (see figure 5 for reference).

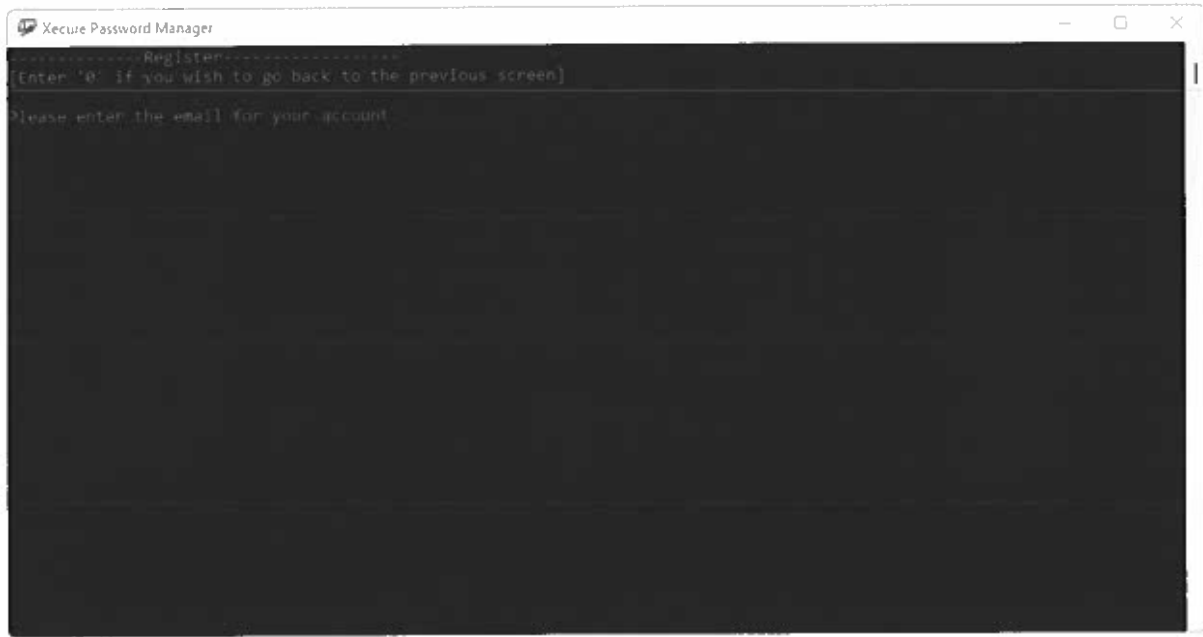


Figure 3: Register Screen



Figure 4: Successful Registration Screen



Figure 5: Login Help Screen

If the user selects the “Forgot Username” selection they will be asked to enter their email address, and if the email address has been registered in the database, the username will be sent to the entered email address (see figure 6 for reference). The user can then use this information to log into the application. On the other hand, if the user forgets their password, they can select the “Forgot Password” selection, and this will allow them to reset their password after verifying their email. In the “Reset Password” screen the user will be asked to enter their email, and if the entered email matches an email that has been registered in the database, a verification code will be sent to that same email address. The user will then have to retrieve that code from their email and enter it into the Xecure Password Manager application. If the entered code matches the code sent to the email, the user will be asked to enter a new password. If the code entered does not match the code sent to the email, the user will be sent back to the main menu, and they will not be able to change their password (see figure 7 for reference).

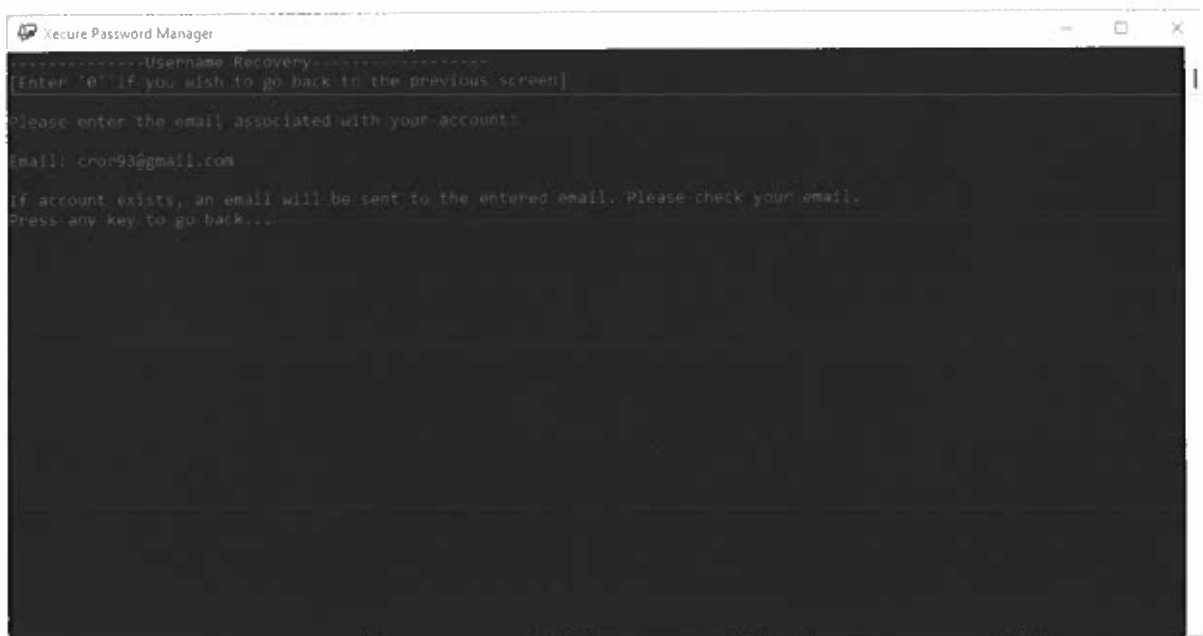


Figure 6: Username Recovery Screen



Figure 7: Reset Password Screen

Once a user logs into the application, they will be greeted with the “Dashboard” menu. From this menu users can select between the options of retrieving a record, adding a new record, updating a record, viewing all records, viewing the record names of all their records, settings, logging out, and exiting the application (see figure 8 for reference). However, if the user has not added any records, they will not be able to use the options of retrieving a record, updating a record, viewing all records, and viewing all record names only. They will instead, see a message letting them know that there are no records in the database, and they will be asked to add a new record in order to use those features.

The “Retrieve a Record” option allows users to retrieve the information of a record they have previously entered. Once this option is selected the user will be asked to enter a record name, and if this record is found, all its information will be displayed to the screen (see figure 9 for reference). Additionally, the password of the record will be copied to the clipboard for fifteen seconds. After fifteen seconds the password will be wiped from the clipboard.



Figure 8: Dashboard Screen

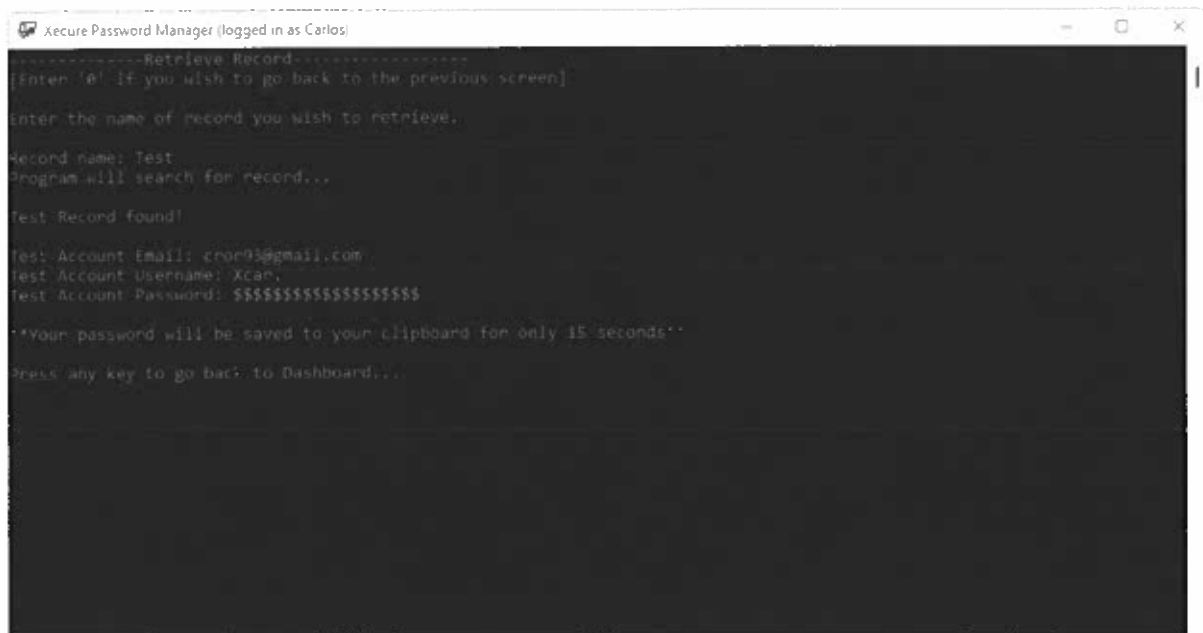


Figure 9: Retrieve Record Screen

To use all the features of the application, users have to add at least one record to their account. This can be done through the “Add a Record” option from the dashboard menu. The “Add Record” menu will prompt the user to enter a record name (see figure 10 for reference). This record name must be unique and cannot be composed of only symbols or numbers. The record name will later be used to retrieve the record. The user will then be prompted to enter a record email and username for the record, but both of these options are optional. Lastly, the user will be asked to enter a password, but the user has the option of generating a random password or entering their own password (see figure 11 for reference). If the user selects the option to generate a new password, a randomly generated password will be displayed to the screen and the user will be asked if he wants to use that password (see figure 12 for reference). If the user is

not happy with this password, they can try another randomly generated password or they can add their own password (see figure 13 for reference).

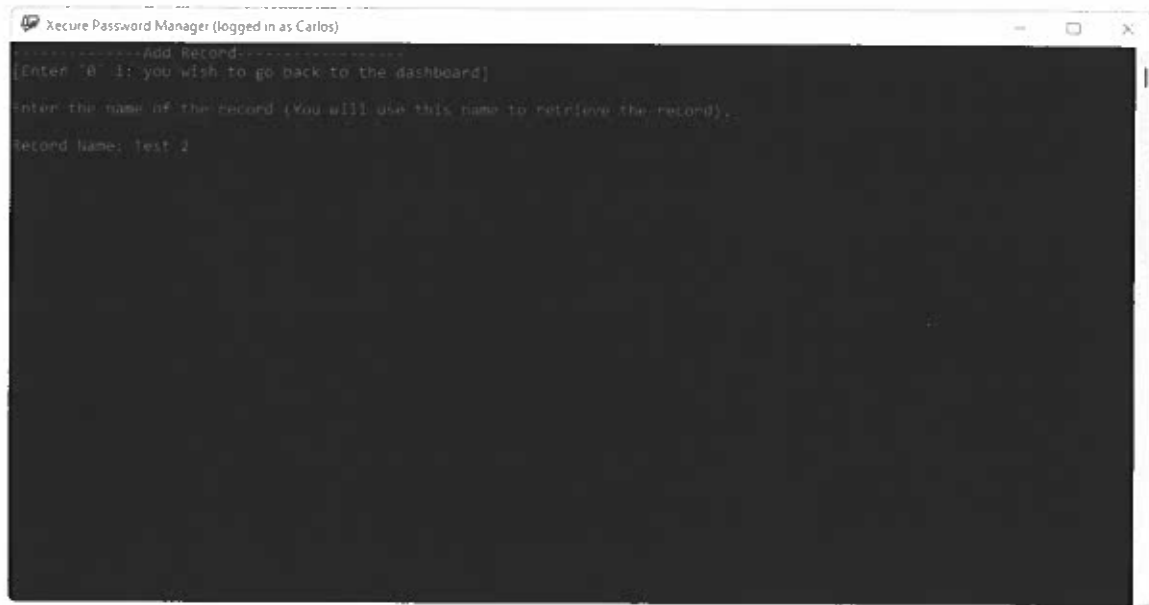


Figure 10: Add Record Screen

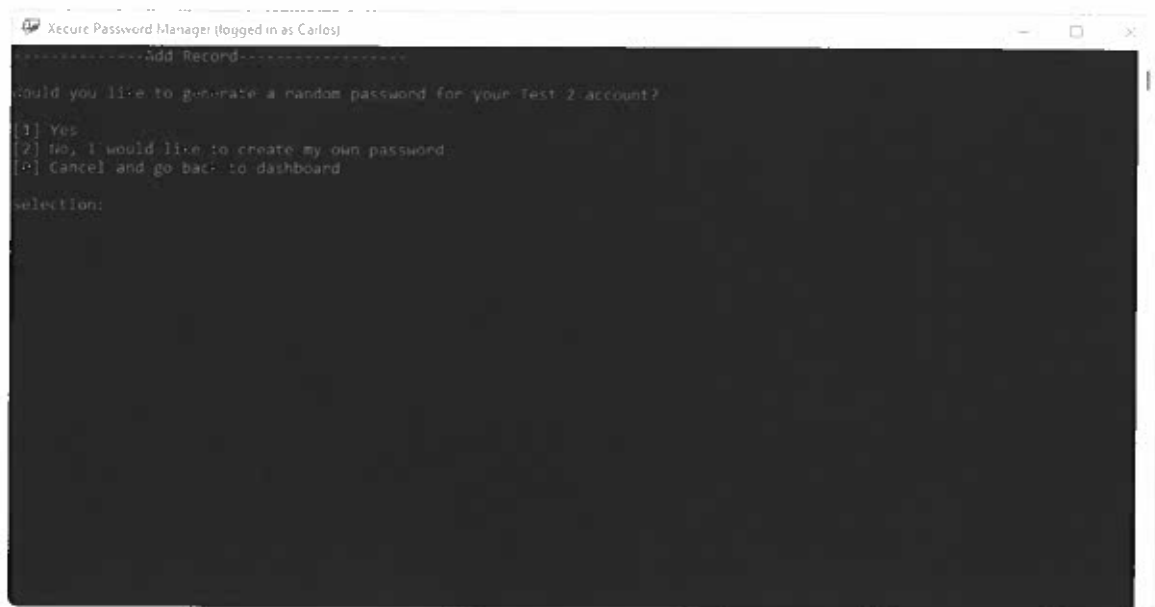


Figure 11: Add Record – Generate Random Password Screen



Figure 12: Generate Random Password Screen



Figure 13: Enter Your Password Screen

Users can edit their records by using the “Update Record” option from the dashboard. Once the user selects the “Update Record” option they will then be prompted to enter the record name of the record they wish to update (see figure 14 for reference). If the record name entered is found within the user’s data, they will be prompted to select what information they want to change about the record (see figure 15 for reference). The user can choose between editing the record’s name, email address, username, or password. Once the user selects an option, they will be asked to enter a new value to update the record to. If the entered input is valid, the record will be updated and the next time the user retrieves that record it will contain the newly updated information (see figure 16 for reference).

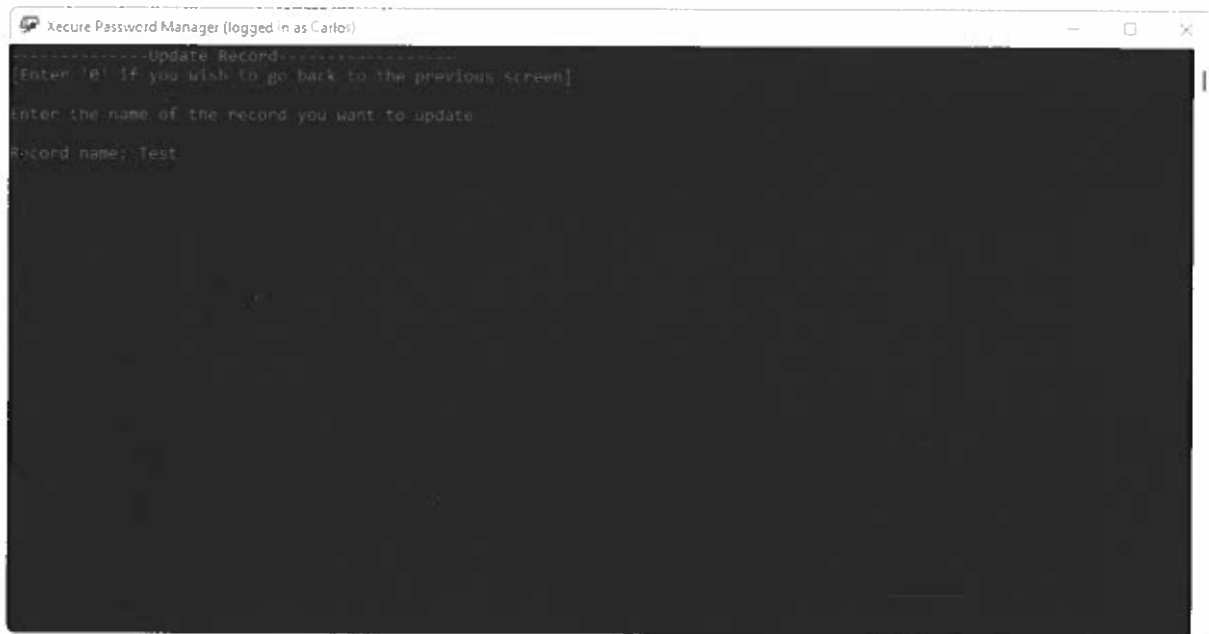


Figure 14: Update Record Screen

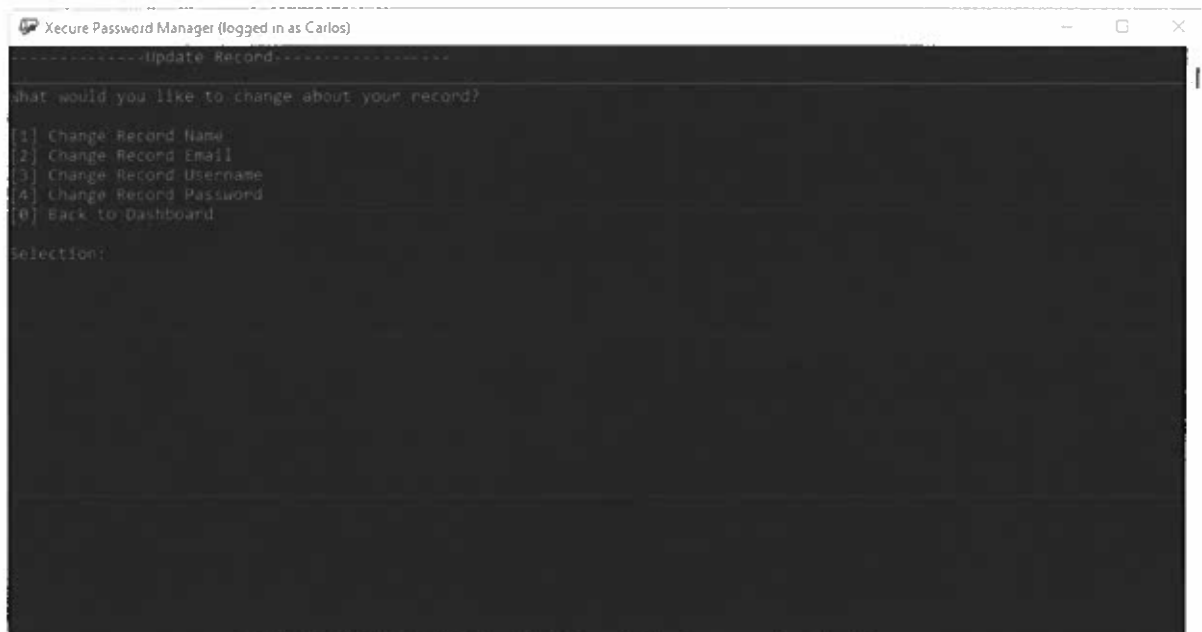


Figure 15: Update Record Screen



Figure 16: Updating Record Name Screen

Users have the option of displaying all their records by using the option of “View All Record” from the dashboard menu. This will display all the records from the currently logged in user, along with all the information stored within them (see figure 17 for reference).



Figure 17: View All Records Screen

Users also have the option of retrieving just the names of their registered records (see figure 18 for reference). This can be used to find a specific record that can then be retrieved individually by the “Retrieve a Record” option from the dashboard. This is a safer option as it will not display the information contained within the record unlike the “View All Records” option.

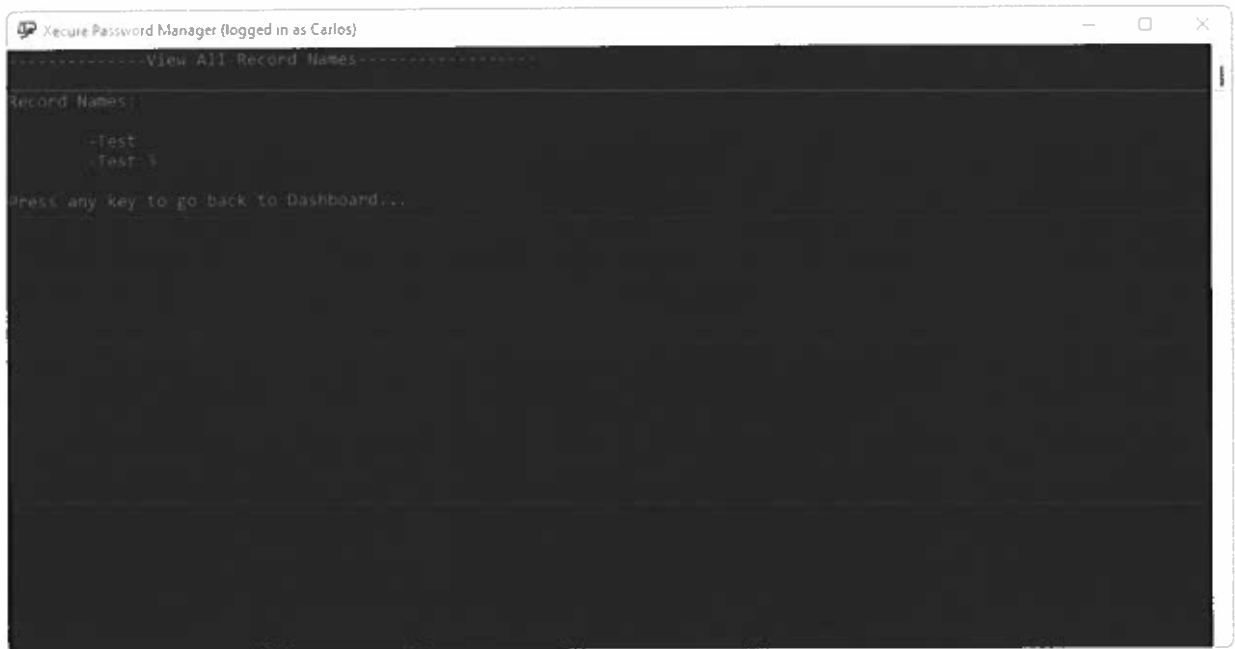


Figure 18: View All Record Names Screen

The application also contains a settings menu that gives users more control over their data within the Xecure Password Manager application. This menu can be accessed by selecting the “Settings & Delete Records” from the dashboard. From the settings menu, users can access the commands to delete a record, delete all records, change master email, or change master password (see figure 19 for reference).



Figure 19: Settings Screen

Users can delete records by selecting between the “Delete a Record”, and “Delete All Records” options from the settings menu. If the user wants to delete a single record, they can use the “Delete a Record” option. This menu will prompt the user to enter the name of the record they wish to delete (see figure 20 for reference). If the record name entered is found within the user’s data, the application will prompt the user to confirm the deletion or cancel this action. If the user confirms the deletion of the record, it will be removed from the database and this data cannot be recovered (see figure 21 and 22 for reference). On the other hand, if the user wishes to delete all their records, they can select the “Delete All Records” from the settings menu. This option deletes all the records contained within the user’s account. Once this option is selected the user will be prompted to confirm the action before deleting the records. If the user confirms the deletion, all the records on the user’s account will be deleted (see figure 23 and 24 for reference).

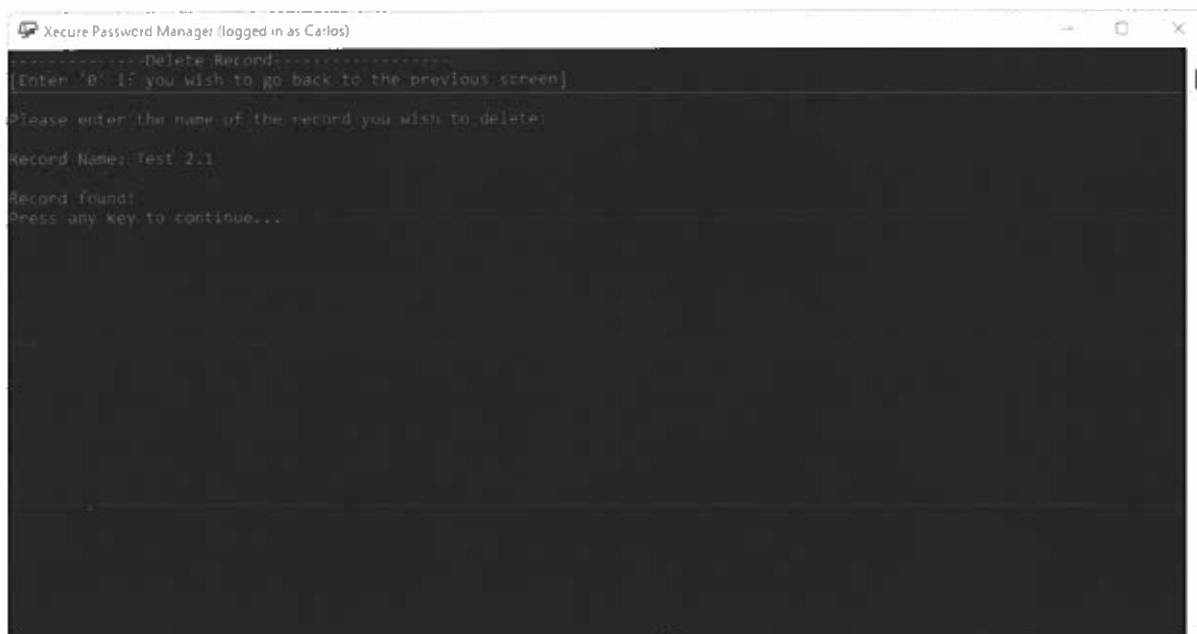


Figure 20: Delete Record Screen

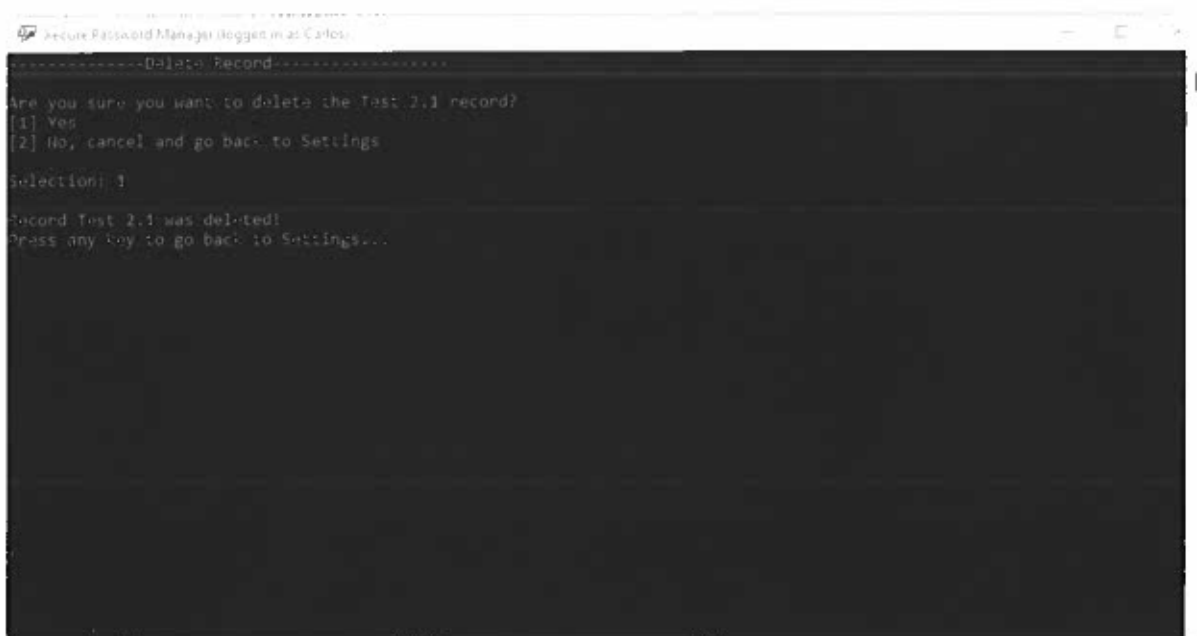


Figure 21: Deleting A Record

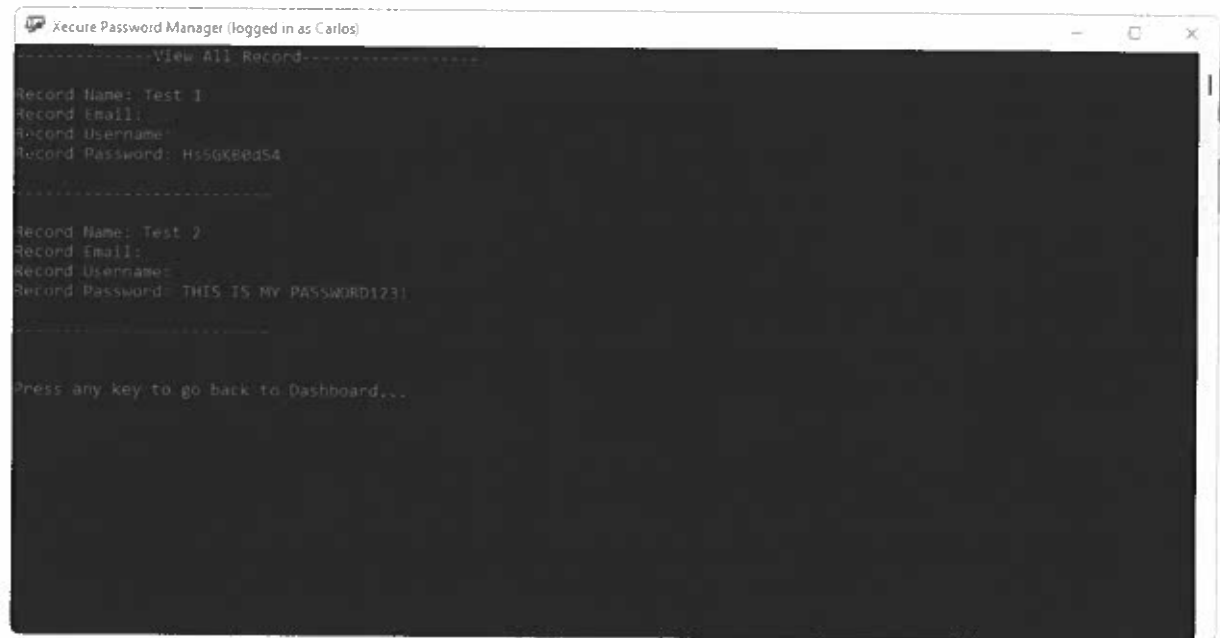


Figure 21: View All Screen After Deleting Record

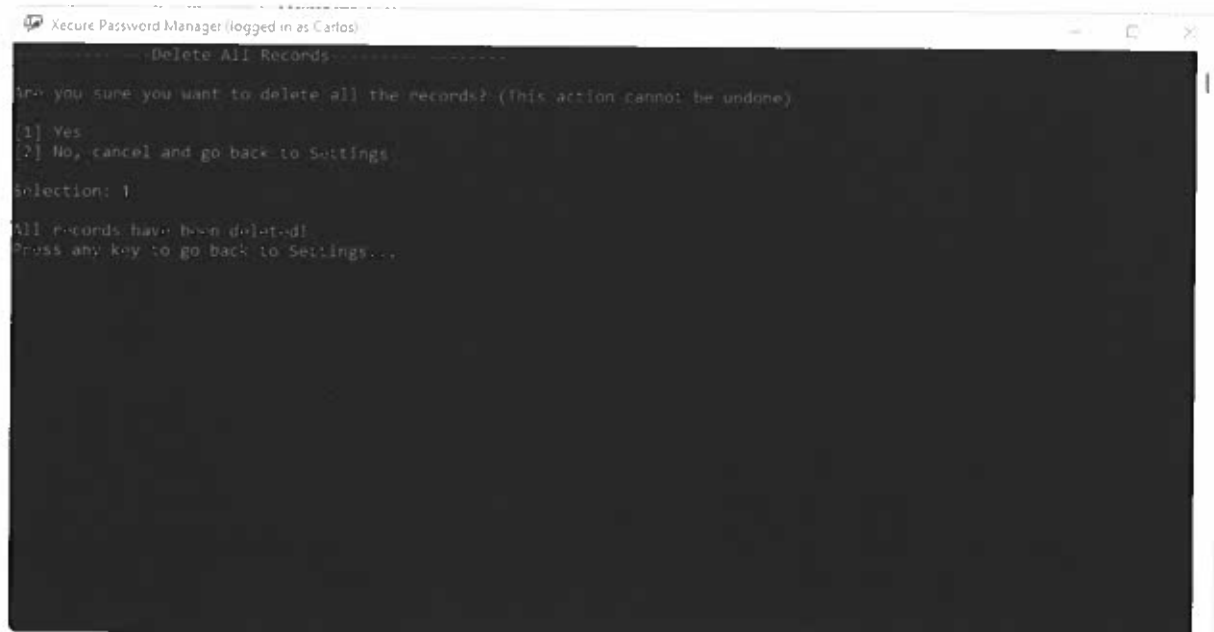


Figure 23: Delete All Records Screen



Figure 24: View All Records After Deleting All Records

User can change the master account email address by selecting the “Change Account Email” option from the settings menu. Once selected, the user will be prompted to enter the email associated with their account (see figure 25 for reference). If the email entered matches the account’s email address, the user will be prompted to check their email and enter a code (see figure 26 for reference). After verifying the entered email, the application sends an email to the user with a verification code that needs to be entered on the Xecure Password Manager Application. If the code entered matched the code that was sent to the user, the user will be prompted to enter a new email address. If the entered input is valid, the user’s master email will be updated, and all future emails will be sent to this new email address (see figure 27 for reference).

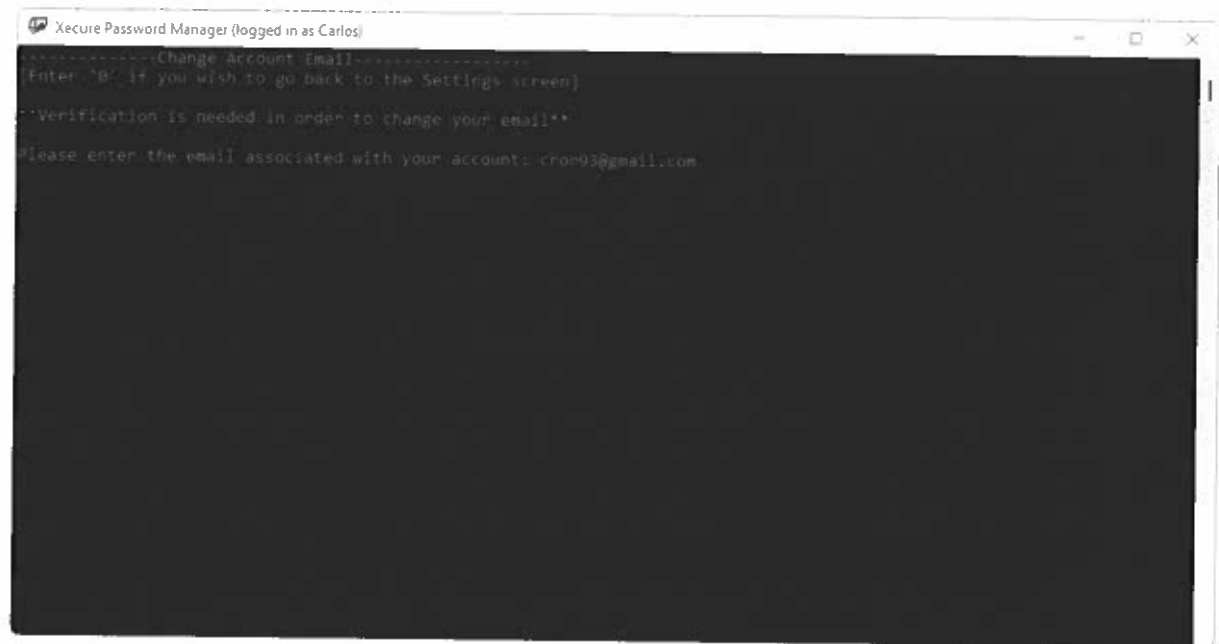


Figure 25: Change Account Email Screen



Figure 26: Change Account Email Screen – Verification Code



Figure 27: Updating Account Email

User can also change the “master” password used to login by selecting the “Change Account Password” option from the settings menu. Once selected, the user will be prompted to enter the email associated with their account (see figure 28 for reference). If the email entered matches the account’s email address, the user will be prompted to check their email and enter a code (see figure 29 for reference). After verifying the entered email, the application sends an email to the user with a verification code that needs to be entered on the Xecure Password Manager Application. If the code entered matched the code that was sent to the user, the user will be prompted to enter a new password for their account. If the password entered is valid, the user’s master password will be updated, and they must use this new password to login (see figure 30 for reference).



Figure 28: Reset Password Screen



Figure 29: Reset Password Screen – Verification Code



Figure 30: Resetting Account Password Screen

The logout menu can be accessed from the “Logout” option on the dashboard, and it can be used to logout the user that is currently logged in without closing the application. Once this option is selected the user will be asked to confirm their selection (see figure 31 for reference). If the user confirms the selection, they will be logged out of their account and they will be taken back to the main menu. Alternatively, if the user wishes to logout and close the application, they can choose the “Exit” option from the dashboard. Once selected the user will be asked to confirm their selection (see figure 32 for reference). If the user confirms the selection, the application will close, and the user will be logged out. Both of these options will wipe the clipboard to ensure that any passwords that were previously saved will not remain after closing the application.

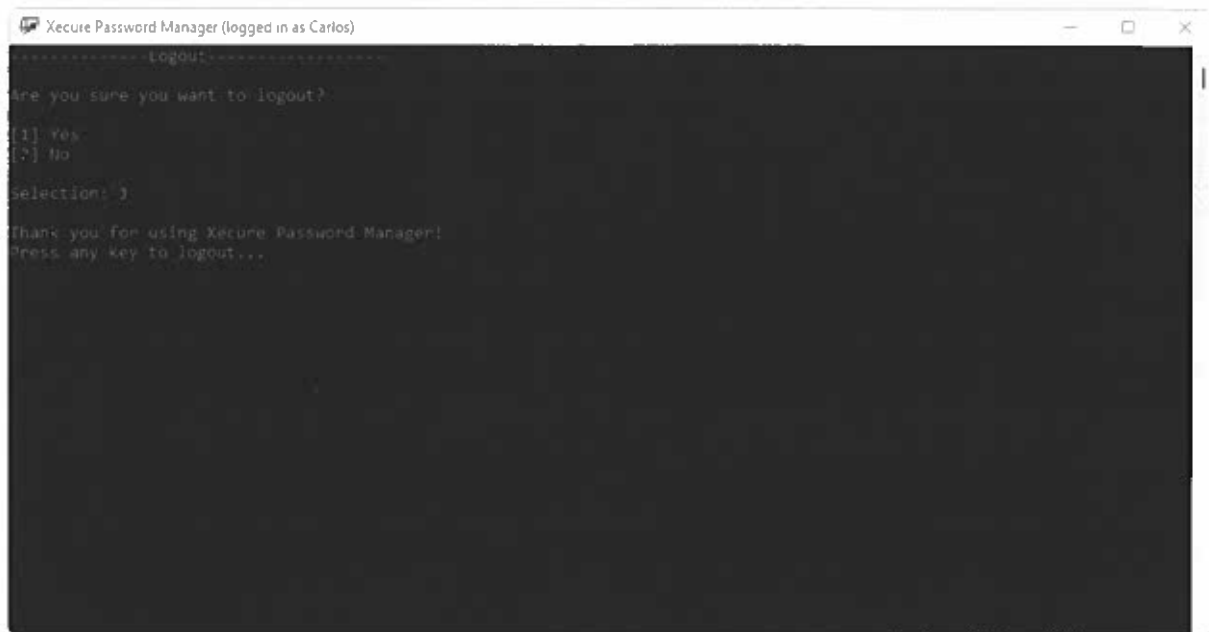


Figure 31: Logout Screen

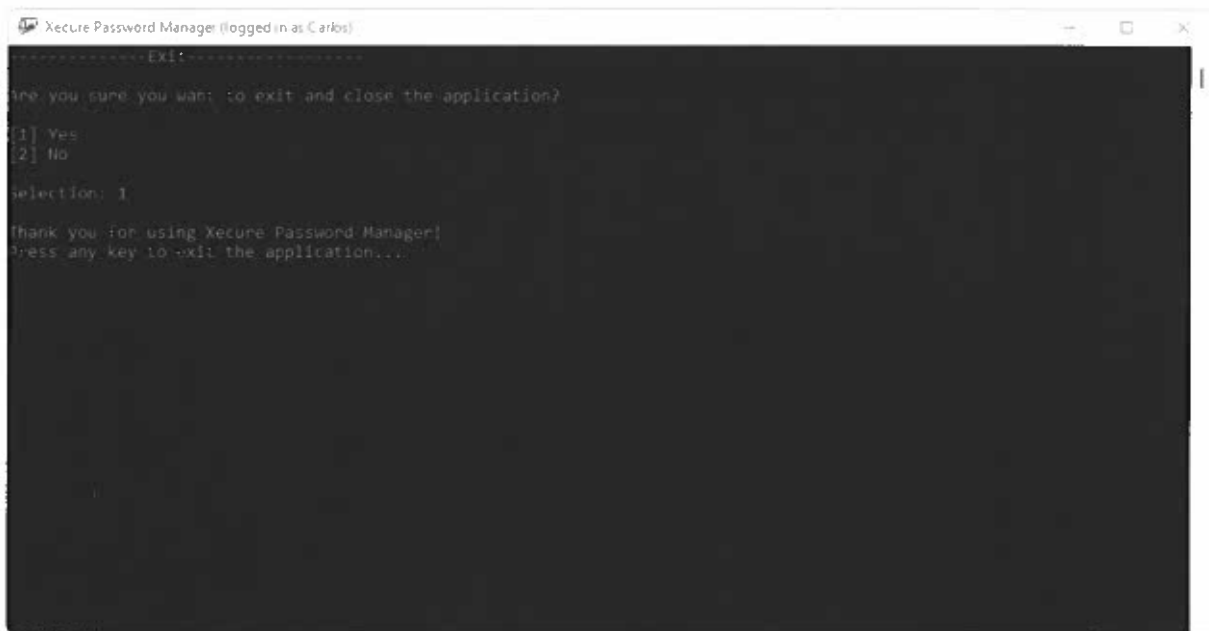


Figure 32: Exit Screen

Various input validation, and error handling methods have been implemented. For example, the menu screen will only accept input that corresponds to the options displayed on the screen. If an invalid input is entered, an error message will be displayed, letting the user know what went wrong (see figure 33 for reference). When attempting to login, carefully constructed error messages are displayed. This is to prevent attackers from getting information about registered accounts (see figure 34 for reference). Inputs that must be unique like usernames, and emails display an error message letting the user know that the entered value is invalid. The application will then prompt the user to enter another value that is unique (see figure 35 for reference). Lastly, creating weak master passwords will result in an error letting the user know the requirements for that input (see figure 36 for reference).



```
Secure Password Manager
-----Main Menu-----
Please select between the following options:
[1] Login
[2] Register
[3] Forgot Login
[4] Exit
Selection: Test
Invalid Input. Please enter a number between 1 and 4.
Press any key to try again...
```

Figure 33: Main Menu Invalid Input



Figure 34: Invalid Login Attempt



Figure 35: Register Screen Email Taken Error

demographic (inexperienced, experience, and very experienced). All three volunteers were asked to perform the following steps:

Tasks:

- 1 – Use the Xecure Password Manager application to register and make an account.
- 2 – Log into the Xecure Password Manager application.
- 3 – Add two new records to your account. Use a randomly generated password for your first record. For your second record, enter your own password.
- 4 – Retrieve and display the information for one of your records.
- 5 – Retrieve and display the information for all of your records.
- 6 – Update the password of one of your records, then retrieve and display it to the screen.
- 7 – Delete one of your records and then display all your records.
- 8 – Change your Master password, then log out and try to login with your new password.
- 9 – Delete all your records, and then try to display all your records to the screen.
- 10 – Exit the application.

Test Results

Major Findings for the Usability Tests:

- 1- Application needed a consistent way to back out of menus.
- 2- Remove User Id information.
- 3- Better description on verification emails needed.
- 4- The Delete Record options difficult to find.
- 5- Email addresses being case sensitive.

6- Verification codes too complicated.

Usability Test Fixes:

- 1- A “back” option was added to every screen, and input prompt.
- 2- Information about user ID, was being displayed but this was corrected.
- 3- Emails now contain a better description of what to do and why the email was sent.
- 4- The “Settings” option is now “Settings & Delete Records”. This will help users find the delete record options.
- 5- Application was updated to remove case sensitive emails.
- 6- Verification codes now limit the number of special characters it uses. Resulting in codes that are easier to read and type.

Major Findings for the Test Cases:

- 1- Application does not sanitize clipboard if user force closes the application.
- 2- No email validation before registration.
- 3- Menu selections accept inputs with leading zeros.
- 4- Cannot properly exit application until the fifteen seconds of the clipboard are done.
- 5- Cannot properly logout of the application until the fifteen seconds of the clipboard are done.

Test Case Fixes:

- 1- Thread handling clipboard functions can now be stopped when exiting application.
- 2- Thread handling clipboard functions can now be stopped when logging out.

The following findings were not corrected in the current version of the application:

- Menu selections accept inputs with leading zeros.
- No email validation before registration.
- Application does not sanitize clipboard if user force closes the application

The previously mentioned findings were not able to be corrected in the current version of the program due to various constraints. These findings were found late in the testing phase and a fix would have required a lot of the testing to be re-done. The “leading zeros” finding does not affect the usability or security of the application and therefore was identified as a low priority. The next update of the application will validate the input to make sure that selections with leading zeros are not accepted. The emails not being validated before registering is a medium severity finding, as it could lead to users signing up with an incorrect email address due to typos. The next update will send a code to the user’s email to verify the email before registration. Lastly, the current structure of the application cannot prevent the user from force closing the application and this can lead to the program closing without sanitizing the clipboard. This bug is tied to the Windows command line, and a solution has not been proposed as of this moment. To minimize this risk, it is advised that users properly exit the application through the dashboard commands.

Challenges Overcome

One of the biggest challenges was the time constraint. Due to financial reasons, I was forced to take the project design and project implementation during the same semester. This was very challenging as I had very little time to research and perform initial tests before I needed to

decide on a project and begin its implementation. This also meant that I had to stick with most of my decisions as I had no time to look for other alternatives. Another big challenge was using the Python language as I had no experience with this programming language. I chose this as my project's programming language because I had always wanted to learn python and using it to create my graduation project would give me the hands-on experience that I desired. However, the biggest challenge came from the encryption and decryption process. I never worked with encryption or decryption algorithms and understanding and applying some of the concepts proved to be very challenging. I had many issues trying to encrypt and decrypt data from the database and this was by far the most time-consuming part of the project. After succeeding in encrypting and decrypting data from the database I had difficulties linking the data to users (since the data was encrypted). The solution to this problem took several weeks, but I was able to address it by giving every user a "user ID" that would be the link between their account and their encrypted data. This user ID is created from a hash that is created from the user's account name and their password. After these challenges were solved, it was only a matter of fixing bugs and testing the application.

Future Enhancements

Due to the nature of Password Manager applications, it is important to maintain a plan to support and update the application. New security vulnerabilities are discovered every day and with the advancements of computing, cracking passwords becomes a much easier task every year. The features below are considered necessary for the success of the application and should be implemented to improve the usability and security of the Xecure application.

Future Updates:

- **Email Validation** – The current version of the application cannot validate that the entered email is a valid and working email. To improve the security of the application users should have to verify their email address before they can register or change their email address.
- **Weak Password Identification** – A feature that identifies weak passwords for user records will be implemented in the future. This feature will display a warning message letting the user know that their password is considered weak.
- **Better Input Validation** – Better input validation will improve the security of the application, but this is a time-consuming task, and it requires continuous monitoring and updating.
- **Better Master Password Policies** – The current version of the application follows the basic guidelines of the NIST 800-63B. However, a stricter password policy should be created for the “master” password. This new policy should prevent users from using well known passwords, and previously used passwords.
- **Confirm Password/Email** – The current version of the application only prompts the user to enter their email/password once. In a future update the user will be asked to re-enter their password/email in order to confirm the input. This will avoid users getting locked out due to typos.
- **Additional Authentication Methods** – Adding other authentication methods will improve the security of the application. Additional authentication methods can include one-time codes sent to the user’s phone, user security questions, and or others.

- **Better Lockout** – The current version of the application only closes the application after six failed login attempts. A feature that locks the account based on the username will be implemented. This will ensure that the locked user cannot login without verifying their account through a code sent to their email address.
- **Reject Leading Zeros** – The current version of the application accepts menu selections that have leading zeros (as long as the input is composed of only leading zeros and a valid integer at the end). A fix will be implemented to validate input that contains leading zeros and reject the input.
- **GUI Version** – A version that utilizes a graphical interface would make it easier to use and should be possible now that the foundation has been implemented.

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- Mavituna, F. (2022, March 14). *SQL Injection Cheat Sheet*. Netsparker. Retrieved from <https://www.invicti.com/blog/web-security/sql-injection-cheat-sheet/>



Test Cases Preview

<https://github.com/Xcar17/Xecure-Password-Manager>

(Please Visit GitHub Repository for a detailed look of the test cases)

Application	Secured Password Manager	Charleston Southern University
PREPARED BY	Carlos Ocasio	Test Cases for Secured Password Manager
REVIEWER	Cybersecurity	
DATE	3/12/2022	

SCENARIO		SCENARIO DESCRIPTION		Testing the Exit screen for unexpected behavior			
TEST CASE ID	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	DIRECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	Enter a valid integer	N/A	1	Application displays exit message and closes program	Application displays exit message and closes program	Pass	
2	Enter a valid integer	N/A	2	Application does not close. User is taken back to previous screen	Application does not close. User is taken back to previous screen	Pass	
3	Enter an invalid integer	N/A	5, 0, -9, -2, 10000	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	
4	Enter an invalid decimal	N/A	10.5, -10000.00	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	
5	Enter an invalid space character	N/A	space char	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	
6	Enter an invalid string	N/A	test, \$aga	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	
7	Enter an invalid long string	N/A	Generating random string	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	
8	Enter no input	N/A	N/A	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	
9	Enter an invalid long integer	N/A	11111111111111111111	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	
10	Enter an invalid symbol character	N/A	4, @, %, &	Error message will let the user know the input is valid	Invalid input message is displayed	Pass	

Application	Keured Password Manager	Charleston Southern University
PREPARED BY	Carlos Ocasio	Test Cases for Keured Password Manager
MAJOR	Cybersecurity	
DATE	2/17/2022	

SCENARIO	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
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Testing the Login screen for unexpected behavior

TEST CASE #	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	Username empty	Registered Users	N/A or Space char	Username cannot be empty. User cannot continue	User cannot continue and "Username cannot be empty" message is displayed	Pass	
2	Unregistered username and empty password	Registered Users	Test and empty	Password cannot be empty. User cannot continue	Invalid login attempt. User cannot continue	Pass	
3	Registered username and empty password	Registered Users	Carlos and empty	Password cannot be empty. User cannot continue	Invalid login attempt. User cannot continue	Pass	
4	Integer username and empty password	Registered Users	123 and empty	Password cannot be empty. User cannot continue	Invalid login attempt. User cannot continue	Pass	
5	Symbols username and empty password	Registered Users	### and empty	Password cannot be empty. User cannot continue	Invalid login attempt. User cannot continue	Pass	
6	Registered username and incorrect password	Registered Users	Carlos and 12345	Invalid login attempt. User cannot continue	Invalid login attempt. User cannot continue	Pass	
7	Registered username and long string password	Registered Users	Carlos and General	Password must be less than 35 characters long	Password must be less than 35 characters long	Pass	
8	Enter an invalid long string as username	Registered Users	Test and General	Username must be less than 35 characters long	Username must be less than 35 characters long	Pass	
9	Incorrect login 6 attempts	Registered Users	N/A	Application should close after 6 failed login attempts	Application closes after 6 failed login attempts	Pass	
10	Enter an invalid long integer as username	Registered Users	1111111111111111	Username must be less than 35 characters long	Username must be less than 35 characters long	Pass	
11	Registered username and invalid long integer as password	Registered Users	Carlos and 11111	Password must be less than 35 characters long	Password must be less than 35 characters long	Pass	
12	Not registered username and invalid long integer as password	Registered Users	Test and 1111111	Password must be less than 35 characters long	Password must be less than 35 characters long	Pass	
13	Attempting to login with password of another user	A Registered User Must Have	Keured and Wp	Invalid login attempt. User cannot continue	Invalid login attempt. User cannot continue	Pass	
14	Symbols username and password	Registered Users	### and ####	Invalid login attempt. User cannot continue	Invalid login attempt. User cannot continue	Pass	
15	Same username and password	Registered Users	Carlos and Carlos	Invalid login attempt. User cannot continue	Invalid login attempt. User cannot continue	Pass	
16	Username as username and password as password	Registered Users	username and password	Invalid login attempt. User cannot continue	Invalid login attempt. User cannot continue	Pass	
17	Admin as username and password	Registered Users	admin and admin	Invalid login attempt. User cannot continue	Invalid login attempt. User cannot continue	Pass	
18	Admin as username and password as password	Registered Users	admin and password	Invalid login attempt. User cannot continue	Invalid login attempt. User cannot continue	Pass	
19	Incorrect login 5 attempts	Registered Users	N/A	Application will close after another failed login attempt	Application displays "Application will close after another failed login attempt"	Pass	
20	Correct login	Registered Users	Carlos and Keured	Application will display "Thank you for logging in Carlos"	"Thank you for logging in Carlos" message displayed and user is taken to main menu		
21	W as username	Registered Users	0	Login cancelled and user is taken back to main menu	Login cancelled and user is taken back to main menu		
22	Registered username and W as password	Registered Users	Carlos and 0	Login cancelled and user is taken back to main menu	Login cancelled and user is taken back to main menu		
23	Username not registered and W as password	Registered Users	Test and 0	Login cancelled and user is taken back to main menu	Login cancelled and user is taken back to main menu		

Application	Account Password Manager	Charleston Southern University
PREPARED BY	Calvin Deaso	Test Cases for Account Password Manager
NAME	Cybersecurity	
DATE	2/17/2022	

SYSTEM	TC Register	Testing the Register screen for unexpected behavior
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Enter Email Prompt				
TEST CASE	TEST CASE DESCRIPTION	PREDICTION	TEST DATA	EXPECTED RESULT
1	Enter an integer	No Registered Users	1 100 783	cannot be composed of only numbers' should be displayed
2	Enter negative number	No Registered Users	-1 -100 -783	Email must be at least 6 characters long displayed
3	Enter decimal number	No Registered Users	15 10.00 75.95	Email must be at least 6 characters long displayed
4	Enter less than 6 characters	No Registered Users	test hello	Email must be at least 6 characters long displayed
5	Enter more than 35 characters	No Registered Users	AAAAAAAAAAAA	Email must be less than 35 characters long displayed
6	Enter letters and integer	No Registered Users	This is a test 1	Email must contain letters, the @ symbol, and a period (.)
7	Enter an invalid space character	No Registered Users	space char	Email cannot be empty
8	Enter @ to back out	No Registered Users	0	User should be taken back to the main menu
9	Enter an invalid long string	No Registered Users	Generating rand	Email must be less than 35 characters long
10	Enter no input	No Registered Users	NA	Email cannot be empty
11	Enter an invalid long integer	No Registered Users	11111111111111111111	Email cannot be composed of only numbers
12	Enter an invalid symbol character	No Registered Users	@ % . / _	Email must be at least 6 characters long displayed
13	Enter an email without a period	No Registered Users	test3@symbol	Email must contain letters, the @ symbol, and a period (.)
14	Enter an email without @ symbol	No Registered Users	test3@gmail.com	Email must contain letters, the @ symbol, and a period (.)
15	Entering email without both period and @ symbol	No Registered Users	test3@gmail	Email must contain letters, the @ symbol, and a period (.)
16	Entering email with both period and @ symbol	No Registered Users	test3@gmail.com	Program accepts input and asks for username
17	Enter an integer	Registered Users	1 100 783	cannot be composed of only numbers' should be displayed
18	Enter negative number	Registered Users	-1 -100 -783	Email must be at least 6 characters long displayed
19	Enter decimal number	Registered Users	15 10.00 75.95	Email must be at least 6 characters long displayed
20	Enter less than 6 characters	Registered Users	test hello	Email must be at least 6 characters long displayed
21	Enter more than 35 characters	Registered Users	AAAAAAAAAAAA	Email must be less than 35 characters long
22	Enter letters and integer	Registered Users	This is a test 1	Email must contain letters, the @ symbol, and a period (.)
23	Enter an invalid space character	Registered Users	space char	Email cannot be empty
24	Enter @ to back out	Registered Users	0	User should be taken back to the main menu
25	Enter an invalid long string	Registered Users	Generating rand	Email must be less than 35 characters long
26	Enter no input	Registered Users	NA	Email cannot be empty
27	Enter an invalid long integer	Registered Users	11111111111111111111	Email cannot be composed of only numbers
28	Enter an invalid symbol character	Registered Users	@ % . / _	Email must be at least 6 characters long displayed
29	Enter an email without a period	Registered Users	test3@symbol	Email must contain letters, the @ symbol, and a period (.)
30	Enter an email without @ symbol	Registered Users	test3@gmail.com	Email must contain letters, the @ symbol, and a period (.)
31	Entering email without both period and @ symbol	Registered Users	test3@gmail	Email must contain letters, the @ symbol, and a period (.)
32	Entering email with both period and @ symbol	Registered Users	test3@gmail.com	Program accepts input and asks for username

Enter Username Prompt				
TEST CASE	TEST CASE DESCRIPTION	PREDICTION	TEST DATA	EXPECTED RESULT
33	Enter an integer	No Registered Users	1 100 783	cannot be composed of only numbers' should be displayed
34	Enter negative number	No Registered Users	-1 -100 -783	Username must be at least 3 characters long
35	Enter decimal number	No Registered Users	15 10.00 75.95	Username must be composed of only letters and numbers. No spaces or special characters
36	Enter less than 3 characters	No Registered Users	My AI	Username must be at least 3 characters long
37	Enter more than 35 characters	No Registered Users	AAAAAAAAAAAA	Username must be less than 35 characters long
38	Enter letters and integer	No Registered Users	This is a test 1	Username must be composed of only letters and numbers. No spaces or special characters
39	Enter an invalid space character	No Registered Users	space char	Username cannot be empty

Application	Account Password Manager	Osweston Southern University Test Cases for Account Password Manager
Prepared By	Saad Oqas O	
Major	Cybersecurity	
Date	31/12/2022	

Scenario: <u>Forgot Login</u>	Scenario Description	Testing the forgot login screen for unexpected behavior
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IT Case	Test Case Description	Prerequisite	Test Data	Expected Result	Actual Result	Status	Comments
1	Enter a valid integer	Registered Users	1	User should be taken to the "Forgot Username" menu	User is taken to the "Forgot Username" menu	Pass	
2	Enter a valid integer	Registered Users	2	User should be taken to the "Forgot Password" menu	User is taken to the "Forgot Password" menu	Pass	
2	Enter a valid integer	Registered Users	0	User should be taken back to the Main menu	User is taken back to the Main menu	Pass	
3	Enter an invalid integer	Registered Users	5, 9, -2, 10000	Please enter a number between 0 and 2	Please enter a number between 0 and 2	Pass	
4	Enter an invalid decimal	Registered Users	10.5, -10000.00	Invalid input message	Invalid input. Please select a number between 0 and 2	Pass	
5	Enter an invalid space character	Registered Users	space char	Invalid input message	Invalid input. Please select a number between 0 and 2	Pass	
6	Enter an invalid string	Registered Users	test	Invalid input message	Invalid input. Please select a number between 0 and 2	Pass	
7	Enter an invalid long string	Registered Users	Generating random	Invalid input message	Invalid input. Please select a number between 0 and 2	Pass	
8	Enter no input	Registered Users	N/A	Invalid input message	Invalid input. Please select a number between 0 and 2	Pass	
9	Enter an invalid long integer	Registered Users	11111111111111111111	Invalid input message	Please enter a number between 0 and 2	Pass	
10	Enter an invalid symbol character	Registered Users	@, #, &, /	Invalid input message	Invalid input. Please select a number between 0 and 2	Pass	

Application	Account Password Manager	
PREPARED BY	Carlos Dario	
VALIDATOR	Cybersecurity	
DATE	20/10/2022	

SUBJECT	TC Add Record	Testing the Add Record screen for unexpected behavior
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Enter Record Name Prompt				
ST/ID	TEST CASE DESCRIPTION	PRECEDITION	TEST DATA	EXPECTED RESULT
1	Enter an integer	No Records in Database	1, 100, 783	Record name cannot be composed of only numbers
2	Enter negative number	No Records in Database	-1, -4, -7	Record name must be at least 3 characters long
3	Enter decimal number longer than 3 characters	No Records in Database	55, 10.00, 75.99	Record names must contain letters. Numbers and special characters are optional
4	Enter decimal number less than 3 characters	No Records in Database	0, 1, 4	Record name must be at least 3 characters long
5	Enter less than 3 characters	No Records in Database	a, a, a, l	Record name must be at least 3 characters long
6	Enter more than 35 characters	No Records in Database	AA	Record name must be less than 35 characters long
7	Enter valid record name	No Records in Database	Test	The program should accept this as a valid record name. The user will then be asked to enter an email for their record
8	Enter 3 letter record name (valid record name)	No Records in Database	hey	The program should accept this as a valid record name. The user will then be asked to enter an email for their record
9	Enter letters and number (valid record name)	No Records in Database	This is a test 1	The program should accept this as a valid record name. The user will then be asked to enter an email for their record
10	Enter an invalid space character	No Records in Database	space char	Record name cannot be empty
11	Enter 'I' to back out	No Records in Database	0	User should be taken back to the Dashboard
12	Enter an invalid long string	No Records in Database	Generating and	Record name must be less than 35 characters long
13	Enter no input	No Records in Database	N/A	Record name cannot be empty
14	Enter an invalid long integer	No Records in Database	0, 1, 4	Record name cannot be composed of only numbers
15	Enter an invalid symbol character	No Records in Database	0, @, %, &	Record name must be at least 3 characters long
16	Enter two valid names in the same input	No Records in Database	Test Test 1	Record name must be at least 3 characters long
17	Enter an integer	Records in Database	1, 100, 783	Record name cannot be composed of only numbers
18	Enter negative number	Records in Database	-1, -4, -7	Record name must be at least 3 characters long
19	Enter decimal number longer than 3 characters	Records in Database	55, 10.00, 75.99	Record names must contain letters. Numbers and special characters are optional
20	Enter decimal number less than 3 characters	Records in Database	0, 1, 4	Record name must be at least 3 characters long
21	Enter less than 3 characters	Records in Database	a, a, a, l	Record name must be at least 3 characters long
22	Enter more than 35 characters	Records in Database	AA	Record name must be less than 35 characters long
23	Enter valid record name	Records in Database	Test	The program should accept this as a valid record name. The user will then be asked to enter an email for their record
24	Enter 3 letter record name (valid record name)	Records in Database	hey	The program should accept this as a valid record name. The user will then be asked to enter an email for their record
25	Enter letters and number (valid record name)	Records in Database	This is a test 1	The program should accept this as a valid record name. The user will then be asked to enter an email for their record
26	Enter an invalid space character	Records in Database	space char	Record name cannot be empty
27	Enter 'I' to back out	Records in Database	0	User should be taken back to the Dashboard
28	Enter an invalid long string	Records in Database	Generating and	Record name must be less than 35 characters long
29	Enter no input	Records in Database	N/A	Record name cannot be empty
30	Enter an invalid long integer	Records in Database	0, 1, 4	Record name cannot be composed of only numbers
31	Enter an invalid symbol character	Records in Database	0, @, %, &	Record name must be at least 3 characters long
32	Entering Record name that belongs to another user	Records in Database	Test	The program should accept this as a valid record name. The user will then be asked to enter an email for their record
33	Entering Record name that is shared by the current user and other user	Records in Database	Test	The program should not accept this as a valid record name. Record name taken, please choose another one
34	Entering a duplicate record name	Records in Database	Test	The program should not accept this as a valid record name. Record name taken, please choose another one
Enter Email Prompt				
ST/ID	TEST CASE DESCRIPTION	PRECEDITION	TEST DATA	EXPECTED RESULT
35	Enter an integer	No Records in Database	1, 100, 783	Email cannot be composed of only numbers should be displayed
36	Enter negative number	No Records in Database	-1, -100, -783	Email must be at least 6 characters long displayed
37	Enter decimal number	No Records in Database	55, 10.00, 75.99	Email must be at least 6 characters long displayed
38	Enter less than 6 characters	No Records in Database	test, hello	Email must be at least 6 characters long displayed
39	Enter more than 35 characters	No Records in Database	AA	Email must be less than 35 characters long
40	Enter letters and number	No Records in Database	This is a test 1	Email must contain letters, the @ symbol, and a period (.)
41	Enter the space character	No Records in Database	space char	This is a valid input and the program should allow the user to continue to the next step
42	Enter 'I' to back out	No Records in Database	0	User should be taken back to the Dashboard
43	Enter an invalid long string	No Records in Database	Generating and	Email must be less than 35 characters long

ST/ID	TEST CASE DESCRIPTION	PRECEDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	Enter an integer	No Records in Database	1, 100, 783	Record name cannot be composed of only numbers	Record name cannot be composed of only numbers	Pass	
2	Enter negative number	No Records in Database	-1, -4, -7	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
3	Enter decimal number longer than 3 characters	No Records in Database	55, 10.00, 75.99	Record names must contain letters. Numbers and special characters are optional	Record names must contain letters. Numbers and special characters are optional	Pass	
4	Enter decimal number less than 3 characters	No Records in Database	0, 1, 4	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
5	Enter less than 3 characters	No Records in Database	a, a, a, l	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
6	Enter more than 35 characters	No Records in Database	AA	Record name must be less than 35 characters long	Record name must be less than 35 characters long	Pass	
7	Enter valid record name	No Records in Database	Test	The program should accept this as a valid record name. The user will then be asked to enter an email for their record	The program accepts the input and adds the user for the record email	Pass	
8	Enter 3 letter record name (valid record name)	No Records in Database	hey	The program should accept this as a valid record name. The user will then be asked to enter an email for their record	The program accepts the input and adds the user for the record email	Pass	
9	Enter letters and number (valid record name)	No Records in Database	This is a test 1	The program should accept this as a valid record name. The user will then be asked to enter an email for their record	The program accepts the input and adds the user for the record email	Pass	
10	Enter an invalid space character	No Records in Database	space char	Record name cannot be empty	Record name cannot be empty	Pass	
11	Enter 'I' to back out	No Records in Database	0	User should be taken back to the Dashboard	User is taken back to the Dashboard	Pass	
12	Enter an invalid long string	No Records in Database	Generating and	Record name must be less than 35 characters long	Record name must be less than 35 characters long	Pass	
13	Enter no input	No Records in Database	N/A	Record name cannot be empty	Record name cannot be empty	Pass	
14	Enter an invalid long integer	No Records in Database	0, 1, 4	Record name cannot be composed of only numbers	Record name cannot be composed of only numbers	Pass	
15	Enter an invalid symbol character	No Records in Database	0, @, %, &	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
16	Enter two valid names in the same input	No Records in Database	Test Test 1	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
17	Enter an integer	Records in Database	1, 100, 783	Record name cannot be composed of only numbers	Record name must be at least 3 characters long	Pass	
18	Enter negative number	Records in Database	-1, -4, -7	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
19	Enter decimal number longer than 3 characters	Records in Database	55, 10.00, 75.99	Record names must contain letters. Numbers and special characters are optional	Record names must contain letters. Numbers and special characters are optional	Pass	
20	Enter decimal number less than 3 characters	Records in Database	0, 1, 4	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
21	Enter less than 3 characters	Records in Database	a, a, a, l	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
22	Enter more than 35 characters	Records in Database	AA	Record name must be less than 35 characters long	Record name must be less than 35 characters long	Pass	
23	Enter valid record name	Records in Database	Test	The program should accept this as a valid record name. The user will then be asked to enter an email for their record	The program accepts the input and adds the user for the record email	Pass	
24	Enter 3 letter record name (valid record name)	Records in Database	hey	The program should accept this as a valid record name. The user will then be asked to enter an email for their record	The program accepts the input and adds the user for the record email	Pass	
25	Enter letters and number (valid record name)	Records in Database	This is a test 1	The program should accept this as a valid record name. The user will then be asked to enter an email for their record	The program accepts the input and adds the user for the record email	Pass	
26	Enter an invalid space character	Records in Database	space char	Record name cannot be empty	Record name cannot be empty	Pass	
27	Enter 'I' to back out	Records in Database	0	User should be taken back to the Dashboard	User is taken back to the Dashboard	Pass	
28	Enter an invalid long string	Records in Database	Generating and	Record name must be less than 35 characters long	Record name must be less than 35 characters long	Pass	
29	Enter no input	Records in Database	N/A	Record name cannot be empty	Record name cannot be empty	Pass	
30	Enter an invalid long integer	Records in Database	0, 1, 4	Record name cannot be composed of only numbers	Record name cannot be composed of only numbers	Pass	
31	Enter an invalid symbol character	Records in Database	0, @, %, &	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
32	Entering Record name that belongs to another user	Records in Database	Test	The program should accept this as a valid record name. The user will then be asked to enter an email for their record	The program accepts the input and adds the user for the record email	Pass	Record name must be unique to the logged in user. No duplicate are
33	Entering Record name that is shared by the current user and other user	Records in Database	Test	The program should not accept this as a valid record name. Record name taken, please choose another one	The program displays the message Record name taken, please choose another one	Pass	Record names must be unique to the logged in user. No duplicate are
34	Entering a duplicate record name	Records in Database	Test	The program should not accept this as a valid record name. Record name taken, please choose another one	The program displays the message Record name taken, please choose another one	Pass	Record names must be unique to the logged in user. No duplicate are

Application	Revised Password Manager			Charleston Southern University
Prepared by	Cariss Ocasio			
Major	Cybersecurity			
Date	2/17/2023			Test Cases for Revised Password Manager

Scenario: TC_Dashboard		Scenario Description		Testing the Dashboard for unexpected behavior			
TEST CASE	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	Enter a valid integer	No Records Registered	1	User should see "No records in database" message	User cannot search for record and a "No records in database" message is displayed	Pass	
2	Enter a valid integer	No Records Registered	2	User should be taken to the Add Record menu	User is taken to the Add Record menu	Pass	
3	Enter a valid integer	No Records Registered	3	see "No records in database" message. Cannot access update	User cannot search for record and a "No records in database" message is displayed	Pass	
4	Enter a valid integer	No Records Registered	4	see "No records in database" message. Cannot access view	User cannot search for record and a "No records in database" message is displayed	Pass	
5	Enter a valid integer	No Records Registered	5	see "No records in database" message. Cannot access delete	User cannot search for record and a "No records in database" message is displayed	Pass	
6	Enter a valid integer	No Records Registered	6	User should be taken to the Settings menu	User is taken to the Settings menu	Pass	
7	Enter a valid integer	No Records Registered	7	User should be taken to the Logout menu	User is taken to the Logout menu	Pass	
8	Enter a valid integer	No Records Registered	8	User should be taken to the Exit menu	User is taken to the Exit menu	Pass	
9	Enter an invalid integer	No Records Registered	0, 9, -2, 10000	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
10	Enter an invalid decimal	No Records Registered	10.5, -10000.00	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
11	Enter an invalid space character	No Records Registered	space char	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
12	Enter an invalid string	No Records Registered	test	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
13	Enter an invalid long string	No Records Registered	Generating random	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
14	Enter no input	No Records Registered	N/A	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
15	Enter an invalid integer	No Records Registered	1111111111111111	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
16	Enter an invalid symbol character	No Records Registered	@ % &	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
17	Enter a valid integer	User Registers Records	1	User should be taken to the Retrieve a Record menu	User is taken to the Retrieve a Record menu	Pass	
18	Enter a valid integer	User Registers Records	2	User should be taken to the Add Record menu	User is taken to the Add Record menu	Pass	
19	Enter a valid integer	User Registers Records	3	User should be taken to the Update Record menu	User is taken to the Update Record menu	Pass	
20	Enter a valid integer	User Registers Records	4	User should be taken to the View All Records menu	User is taken to the View All Records menu	Pass	
21	Enter a valid integer	User Registers Records	5	User should be taken to the View All Record Names Only menu	User is taken to the View All Record Names Only menu	Pass	
22	Enter a valid integer	User Registers Records	6	User should be taken to the Settings menu	User is taken to the Settings menu	Pass	
23	Enter a valid integer	User Registers Records	7	User should be taken to the Logout menu	User is taken to the Logout menu	Pass	
24	Enter a valid integer	User Registers Records	8	User should be taken to the Exit menu	User is taken to the Exit menu	Pass	
25	Enter an invalid integer	User Registers Records	0, 9, -2, 10000	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
26	Enter an invalid decimal	User Registers Records	10.5, -10000.00	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
27	Enter an invalid space character	User Registers Records	space char	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
28	Enter an invalid string	User Registers Records	test	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
29	Enter an invalid long string	User Registers Records	Generating random	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
30	Enter no input	User Registers Records	N/A	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
31	Enter an invalid long integer	User Registers Records	1111111111111111	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	
32	Enter an invalid symbol character	User Registers Records	@ % &	Error message will be the user know the input is invalid	"Please enter a number between 1 and 5" displayed	Pass	

Application	NeuroRx Research Program	Citation South Test Cases for NeuroRx
PREPARED BY	Cathy O'Casey	
MAJOR	Cybersecurity	
DATE	2/17/2022	

SCENARIO ID	UPDATE RECORD	SCENARIO DESCRIPTION
		Testing the Update Record screen for unexpected behavior

Record Name Prompt					
Sl CASE	TEST CASE DESCRIPTION	PREDICTION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT
1)	Accessing system with no records in database	No Records in Database	N/A	No records in database message should be displayed. User should not be able to continue	User is not able to continue. No records in database message displayed
2)	Enter an integer	Records in Database	1, 100, 783	Record name cannot be composed of only numbers	Record name cannot be composed of only numbers
3)	Enter negative number	Records in Database	-1, -0, -7	Record name must be at least 3 characters long	Record name must be at least 3 characters long
4)	Enter decimal number longer than 3 characters	Records in Database	55.0, 0.0, 75.5	Record names must contain letters. Numbers and special characters are optional. Special characters allowed: Underscores (_), spaces (_), and periods (_)	Record names must contain letters. Numbers and special characters are optional. Special characters allowed: Underscores (_), spaces (_), and periods (_)
5)	Enter decimal number less than 3 characters	Records in Database	0, 1, 4	Record name must be at least 3 characters long	Record name must be at least 3 characters long
6)	Enter less than 35 characters	Records in Database	a aa al	Record name must be at least 3 characters long	Record name must be at least 3 characters long
7)	Enter more than 35 characters	Records in Database	AAAAAAAAAAAA	Record name must be less than 35 characters long	Record name must be less than 35 characters long
8)	Enter letters and digits (Test ID) recording in database	Records in Database	Test ID00	Should not find record since it not in database. Test ID00 record not found, displayed	Record not found
9)	Enter an invalid space character	Records in Database	space char	Record name cannot be empty	Record name cannot be empty
10)	Enter _ to back out	Records in Database	_	User should be taken back to the Dashboard	User is taken back to the Dashboard
11)	Enter an invalid long string	Records in Database	Generating and	Record name must be less than 35 characters long	Record name must be less than 35 characters long
12)	Enter an input	Records in Database	WA	Record name cannot be empty	Record name cannot be empty
13)	Enter an invalid long integer	Records in Database	mmmmmmmmmm	Record name cannot be composed of only numbers	Record name cannot be composed of only numbers
14)	Enter an invalid symbol character	Records in Database	@, %, &	Record name must be at least 3 characters long	Record name must be at least 3 characters long
15)	Enter a valid record name that has been saved to database	Records in Database	Test	Program should find record and display the record menu	Program found the record and displayed the record menu
16)	Enter a valid record name that has not been saved to database	Records in Database	Invalid	No record will be found and a message will let the user know that no record has been found	Record not found
17)	Enter two valid names in the same input	Records in Database	Test Test	No record will be found and a message will let the user know that no record has been found	Record not found
18)	Entering Record name that belongs only to another user	Records in Database	Twitter	No record will be found and a message will let the user know that no record has been found	Record not found
19)	Entering Record name that is shared by the current user and other	Records in Database	Facebook	Program should find the record that corresponds to the user logged in	Program finds the record of the user logged in and displays the record menu

Record Update Options Menu Tests							
ST CASE	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
20	Create a valid member	Records n Database	1	User should be taken to the Change Record Name screen	User is taken to the Change Record Name screen	Pass	
21	Create a valid member	Records n Database	2	User should be taken to the Change Record Email screen	User is taken to the Change Record Email screen	Pass	
22	Create a valid member	Records n Database	3	User should be taken to the Change Record Username screen	User is taken to the Change Record Username screen	Pass	
23	Create a valid member	Records n Database	4	User should be taken to the Change Record Password screen	User is taken to the Change Record Password screen	Pass	
24	Create a valid member	Records n Database	0	The user should be taken back to the Dashboard	The user is taken back to the Dashboard	Pass	
25	Create an invalid member	Records n Database	5, 3, -2, 0000	Passes enter a number between 0 and 4 displayed	Passes enter a number between 0 and 4 displayed	Pass	
26	Create an invalid decimal	Records n Database	10 5, -10000, 0	Invalid input. Passes enter a number between 0 and 4 displayed	Invalid input. Passes enter a number between 0 and 4 displayed	Pass	
27	Create an invalid space character	Records n Database	space char	Invalid input. Passes enter a number between 0 and 4 displayed	Invalid input. Passes enter a number between 0 and 4 displayed	Pass	
28	Create an invalid string	Records n Database	test	Invalid input. Passes enter a number between 0 and 4 displayed	Invalid input. Passes enter a number between 0 and 4 displayed	Pass	
29	Create an invalid long string	Records n Database	Enter any long	Invalid input. Passes enter a number between 0 and 4 displayed	Invalid input. Passes enter a number between 0 and 4 displayed	Pass	
30	Create an invalid output	Records n Database	NA	Invalid input. Passes enter a number between 0 and 4 displayed	Invalid input. Passes enter a number between 0 and 4 displayed	Pass	
31	Create an invalid member	Records n Database	mmmmmmmm	Passes enter a number between 0 and 4 displayed	Invalid input. Passes enter a number between 0 and 4 displayed	Pass	
32	Create an invalid symbol character	Records n Database	@ @ % % . .	Invalid input. Passes enter a number between 0 and 4 displayed	Invalid input. Passes enter a number between 0 and 4 displayed	Pass	
New Record Name Prompt							
ST CASE	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
33	Create an integer	Records n Database	1, 100, 783	Record name cannot be composed of only numbers	Record name cannot be composed of only numbers	Pass	
34	Create an integer number	Records n Database	-1, 0, -7	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
35	Create decimal number longer than 3 characters	Records n Database	55, 10.00, 75.3	Record names must contain letters. Numbers and special characters are optional. Special characters allowed: Underscore (_), space (), and periods (.)	Record names must contain letters. Numbers and special characters are optional. Special characters allowed: Underscore (_), space (), and periods (.)	Pass	
36	Create decimal number less than 3 characters	Records n Database	0, 1, 4	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	
37	Create less than 3 characters	Records n Database	a aa x!	Record name must be at least 3 characters long	Record name must be at least 3 characters long	Pass	

Application	Record Password Manager	Charleson Scott Test Case for Record Password Manager
Prepared By	Charleson Scott	
Version	1.0	
Date	2020-07-20	

Test Case ID	TC-001	Test Case Description	Testing the View All Record Names Only screen for unexpected behavior
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Test Case	Test Case Description	Precondition	Test Data	Expected Result	Actual Result	Status	Comments
1	Accessing screen with no records in database	No Records in Database	N/A	No records in database message should be displayed. User should not be able to see any record names	User is not able to continue. No records in database message displayed	Pass	
2	Other users have records but logged in user has no records in database	User has no records in database	N/A	No records in database message should be displayed. User should not be able to see any record names	No records in database message displayed	Pass	
3	User has a record in database	User Record in Database	N/A	Record name should be displayed to the screen	Record name is displayed to the screen	Pass	
4	User has more than one record in database	User Records in Database	N/A	The names for all of the records of the currently logged in user should be displayed to the screen	The names for all of the records of the currently logged in user are displayed to the screen	Pass	
5	User has no records but other users have multiple records in database	User has no records in database	N/A	No records in database message should be displayed. User should not be able to see any record names	No records in database message displayed	Pass	
6	A record name is the same as the record name of another user	User has no records in database	N/A	Only the record names of the current user should be displayed	Only the user's record names are displayed	Pass	

Application	Netcard Password Manager			Developer: Southern Union Test Cases for Netcard Password Manager
PREPARED BY	Carlos Desazo			
MAJOR	Cybersecurity			
DATE	2/7/2022			

STANDARD: TC: View All Records	SCENARIO DESCRIPTION	Testing the View All Records screen for unexpected behavior		
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ST CASE	TEST CASE DESCRIPTION	PREREQUISITE	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	Accessing screen with no records in database	No Records in Database	N/A	No records in database message should be displayed. User should not be able to see any records	User is not able to continue. No records in database message displayed	Pass	
2	Other users have records but logged in user has no records in database	User has no records in database	N/A	No records in database message should be displayed. User should not be able to see any records	No records in database message displayed	Pass	
3	User has a record in database	User Record in Database	N/A	Record information should be displayed to the screen	Record information is displayed to the screen	Pass	
4	User has more than one record in database	User Record in Database	N/A	The information for all of the records should be displayed to the screen	The information for all of the records is displayed to the screen	Pass	
5	User has no records but other users have multiple records in database	User has no records in database	N/A	No records in database message should be displayed. User should not be able to see any records	No records in database message displayed	Pass	
6	A record name is the same as the the record name of another user	User has no records in database	N/A	Only the records of the current user should be displayed	Only the user's records are displayed	Pass	

Application	Secured Passes of the system			Darkroom 36
PREPARED BY	Cedric Desros			Test Cases for Manager
MAKER	Cedric Desros			
DATE	2017/02/22			

SCENARIO TC_Change_Account_Password

SCENARIO DESCRIPTION

Testing the Change Account Password screen for unexpected behavior

ST CASE	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	Enter an empty	No Records in Database	1, 00, 783	Email cannot be composed of only numbers displayed	Email cannot be composed of only numbers	Pass	
2	Enter negative number	No Records in Database	-1, -700, -783	Email must be at least 6 characters long displayed	Email must be at least 6 characters long	Pass	
3	Enter decimal number	No Records in Database	15, 10.00, 75.56	Email must be at least 6 characters long displayed	Email must be at least 6 characters long	Pass	
4	Enter less than 6 characters	No Records in Database	ver, 1mb0	Email must be at least 6 characters long displayed	Email must be at least 6 characters long	Pass	
5	Enter more than 35 characters	No Records in Database	AAAAAAAAAAAAAA	Email must be less than 35 characters long	Email must be less than 35 characters long	Pass	
6	Enter numbers and empty	No Records in Database	This is a test 1	Email must contain letters, the @ symbol, and a period (.)	Email cannot be empty	Pass	
7	Enter an invalid special character	No Records in Database	space chr	Email cannot be empty displayed	User is taken back to the settings menu	Pass	
8	Enter '!' to block out	No Records in Database	0	User should be taken back to the settings menu	Email must be less than 35 characters long	Pass	
9	Enter an invalid long string	No Records in Database	Generate any long	Email must be less than 35 characters long	Email cannot be empty	Pass	
10	Enter no input	No Records in Database	N/A	Email cannot be empty	Email cannot be empty	Pass	
11	Enter an invalid long string	No Records in Database	Generate any long	Email cannot be composed of only numbers	Email cannot be composed of only numbers	Pass	
12	Enter an invalid special character	No Records in Database	@, %, &	Email must be at least 6 characters long displayed	Email must be at least 6 characters long displayed	Pass	
13	Enter an invalid symbol	No Records in Database	on 500g gummy	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
14	Enter an email without period	No Records in Database	on 500g gummy	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
15	Enter an email without @ symbol	No Records in Database	on 500g gummy	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
16	Enter an email without both period and @ symbol (not registered)	No Records in Database	on 500g gummy	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
17	Enter an email with both period and @ symbol (registered and same as log)	No Records in Database	on 500g gummy	The email@gmail.com email is not valid. User should not continue	The email@gmail.com email is not valid. User is not able to continue	Pass	
18	Enter an email with both period and @ symbol (registered email but not the same as log)	No Records in Database	on 500g gummy	The email@gmail.com email is not valid. User should not continue	The email@gmail.com email is not valid. User cannot continue	Pass	User should not be able to change the email to someone else's account
19	Enter an empty	Records in Database	1, 00, 783	Email cannot be composed of only numbers displayed	Email cannot be composed of only numbers	Pass	
20	Enter negative number	Records in Database	-1, -700, -783	Email must be at least 6 characters long displayed	Email must be at least 6 characters long	Pass	
21	Enter decimal number	Records in Database	15, 10.00, 75.56	Email must be at least 6 characters long displayed	Email must be at least 6 characters long	Pass	
22	Enter less than 6 characters	Records in Database	ver, 1mb0	Email must be at least 6 characters long displayed	Email must be at least 6 characters long	Pass	
23	Enter more than 35 characters	Records in Database	AAAAAAAAAAAAAA	Email must be less than 35 characters long	Email must be less than 35 characters long	Pass	
24	Enter letters and empty	Records in Database	This is a test 1	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
25	Enter an invalid special character	Records in Database	space chr	Email cannot be empty displayed	Email cannot be empty	Pass	
26	Enter '!' to block out	Records in Database	0	User should be taken back to the settings menu	User is taken back to the settings menu	Pass	
27	Enter an invalid long string	Records in Database	Generate any long	Email must be less than 35 characters long	Email must be less than 35 characters long	Pass	
28	Enter no input	Records in Database	N/A	Email cannot be empty	Email cannot be empty	Pass	
29	Enter an invalid long string	Records in Database	Generate any long	Email cannot be composed of only numbers	Email cannot be composed of only numbers	Pass	
30	Enter an invalid symbol	Records in Database	@, %, &	Email must be at least 6 characters long displayed	Email must be at least 6 characters long displayed	Pass	
31	Enter an email without period	Records in Database	on 500g gummy	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
32	Enter an email without @ symbol	Records in Database	on 500g gummy	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
33	Enter an email without both period and @ symbol	Records in Database	on 500g gummy	Email must contain letters, the @ symbol, and a period (.)	Email must contain letters, the @ symbol, and a period (.)	Pass	
34	Enter an email with both period and @ symbol (not registered)	Records in Database	on 500g gummy	The email@gmail.com email is not valid. User should not continue	The email@gmail.com email is not valid. User is not able to continue	Pass	
35	Enter an email with both period and @ symbol (registered and same as log)	Records in Database	on 500g gummy	The email@gmail.com email is not valid. User should not continue	The email@gmail.com email is not valid. User cannot continue	Pass	User should not be able to change the email to someone else's account
36	Enter an email with both period and @ symbol (registered email but not the same as log)	Records in Database	on 500g gummy	The email@gmail.com email is not valid. User should not continue	The email@gmail.com email is not valid. User cannot continue	Pass	
Enter Verification Code Prompt							
37	Enter an empty	No Records in Database	1, 00, 783	User could not be verified. User cannot continue and should be taken back to the match. The account could not be verified. User cannot continue and is taken back to Change Account Password screen	User could not be verified. User cannot continue and is taken back to Change Account Password screen	Pass	
38	Enter negative number	No Records in Database	-1, -9, -3	User could not be verified. User cannot continue and should be taken back to the match. The account could not be verified. User cannot continue and is taken back to Change Account Password screen	User could not be verified. User cannot continue and is taken back to Change Account Password screen	Pass	
39	Enter decimal number	No Records in Database	15, 10.00, 75.56	User could not be verified. User cannot continue and should be taken back to the match. The account could not be verified. User cannot continue and is taken back to Change Account Password screen	User could not be verified. User cannot continue and is taken back to Change Account Password screen	Pass	
40	Enter less than 3 characters	No Records in Database	ver, 1mb0	User could not be verified. User cannot continue and should be taken back to the match. The account could not be verified. User cannot continue and is taken back to Change Account Password screen	User could not be verified. User cannot continue and is taken back to Change Account Password screen	Pass	
41	Enter more than 35 characters	No Records in Database	AAAAAAAAAAAAAA	User could not be verified. User cannot continue and should be taken back to the match. The account could not be verified. User cannot continue and is taken back to Change Account Password screen	User could not be verified. User cannot continue and is taken back to Change Account Password screen	Pass	

Application	Xecured Password Manager	Defecters Team Test Cases for Xecured
PREPARED BY	Carlos Ocaso	
MODULE	CyberSecurity	
DATE	2/17/2022	

Scenario: TC_Delete_All Records Testing the Delete All Records screen for unexpected behavior

Are you sure you want to delete all the records? Prompt							
ST CASE	TEST CASE DESCRIPTION	PREREQUISITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	No records in database	No Records in Database	N/A	No records in database message should be displayed. User should not be able to continue	User is not able to continue. No records in database message displayed	Pass	
2	Enter a valid integer	Records in Database	1	Application should delete all the record and will display a message letting the user know	All records have been deleted displayed and all of the user's records are deleted	Pass	
3	Enter a valid integer	Records in Database	2	Application should not delete any records and the user will be taken back to the Settings menu	Records are not deleted and the user is taken back to the Settings menu. The records existed. Display	Pass	
4	Enter an invalid integer	Records in Database	5, 0, 9, -2, 000	Error message will let the user know the input is valid	Invalid Input. Please enter a number between 1 and 2	Pass	
5	Enter an invalid decimal	Records in Database	10.5, -1000.00	Error message will let the user know the input is valid	Invalid Input. Please enter a number between 1 and 2	Pass	
6	Enter an invalid space character	Records in Database	space char	Error message will let the user know the input is valid	Invalid Input. Please enter a number between 1 and 2	Pass	
7	Enter an invalid string	Records in Database	yes, Sdfgh, a	Error message will let the user know the input is valid	Invalid Input. Please enter a number between 1 and 2	Pass	
8	Enter an invalid long string	Records in Database	Generating rand	Error message will let the user know the input is valid	Invalid Input. Please enter a number between 1 and 2	Pass	
9	Enter no input	Records in Database	N/A	Error message will let the user know the input is valid	Selection cannot be blank	Pass	
10	Enter an invalid long integer	Records in Database	TTTTTTTTTTTTTTTT	Error message will let the user know the input is valid	Invalid Input. Please enter a number between 1 and 2	Pass	
11	Enter an invalid symbol character	Records in Database	@ % ^ &	Error message will let the user know the input is valid	Invalid Input. Please enter a number between 1 and 2	Pass	

Application	Account Password Manager	Charleston Southern University
PREPARED BY	Carlo Ducio	Test Cases for Account Password Manager
MAJOR	Cybersecurity	
DATE	2/17/2022	

Scenario: TC_User_Settings		Testing the User Settings for unexpected behavior					
STC#	TEST CASE DESCRIPTION	PRECONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENTS
1	Enter a valid integer	No Records in Database	1	User is taken to the Delete a Record screen with a "No records in Database" message. User is taken to the Delete All Records screen with a "No records in Database" message. User is taken to the Change Master Account Email screen.	Delete a Record screen with a "No records in Database" message. User is taken to the Delete All Records screen with a "No records in Database" message. User is taken to the Change Master Account Email screen.	Pass	
2	Enter a valid integer	No Records in Database	2	User should be taken to the Change Master Account Password screen.	User is taken to the Change Master Account Password screen.	Pass	
3	Enter a valid integer	No Records in Database	3	The user should be taken back to the Dashboard.	The user is taken back to the Dashboard.	Pass	
4	Enter a valid integer	No Records in Database	4	Please enter a number between 0 and 4 displayed.	Please enter a number between 0 and 4 displayed.	Pass	
5	Enter an invalid integer	No Records in Database	5, 9, -2, 10000	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
6	Enter an invalid decimal	No Records in Database	10.5, -10000.00	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
7	Enter an invalid space character	No Records in Database	space char	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
8	Enter an invalid string	No Records in Database	test	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
9	Enter an invalid long string	No Records in Database	Generating and	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
10	Enter no input	No Records in Database	N/A	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
11	Enter an invalid long integer	No Records in Database	11111111111111111111	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
12	Enter an invalid symbol character	No Records in Database	@, %, %, %	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
13	Enter a valid integer	Records in Database	1	User should be taken to the Delete a Record screen.	User is taken to the Delete a Record screen.	Pass	
14	Enter a valid integer	Records in Database	2	User should be taken to the Delete All Records screen.	User is taken to the Delete All Records screen.	Pass	
15	Enter a valid integer	Records in Database	3	User should be taken to the Change Master Account Email screen.	User is taken to the Change Master Account Email screen.	Pass	
16	Enter a valid integer	Records in Database	4	User should be taken to the Change Master Account Password screen.	User is taken to the Change Master Account Password screen.	Pass	
17	Enter a valid integer	Records in Database	0	The user should be taken back to the Dashboard.	The user is taken back to the Dashboard.	Pass	
18	Enter an invalid integer	Records in Database	5, 9, -2, 10000	Please enter a number between 0 and 4 displayed.	Please enter a number between 0 and 4 displayed.	Pass	
19	Enter an invalid decimal	Records in Database	10.5, -10000.00	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
20	Enter an invalid space character	Records in Database	space char	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
21	Enter an invalid string	Records in Database	test	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
22	Enter an invalid long string	Records in Database	Generating and	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
23	Enter no input	Records in Database	N/A	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
24	Enter an invalid long integer	Records in Database	11111111111111111111	Please enter a number between 0 and 4 displayed.	Please enter a number between 0 and 4 displayed.	Pass	
25	Enter an invalid symbol character	Records in Database	@, %, %, %	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	
26	Enter an invalid symbol character	Records in Database	@, %, %, %	Invalid input. Please enter a number between 0 and 4 displayed.	Invalid input. Please enter a number between 0 and 4 displayed.	Pass	



Xecure

SECURE PASSWORD MANAGER

Date: April 20, 2022

Student: Carlos R. Ocasio

Major: Cybersecurity

Course: CSCI 499 Senior Project Defense

Advisor: Prof. Henderson

INTRODUCTION

- Management of passwords is one of the most important aspects of Cybersecurity.
- Strong and unique passwords are vital for account security.
- Most users have far too many online accounts to remember a unique password for each account.
- The average user has over 90 different online accounts.

PROBLEM

- A Google survey found that 52% of users reuse the same password for multiple accounts.
- There is a limit to how many passwords a user can remember.
- Reusing passwords:
 - Leaves users vulnerable to intrusion attacks.
 - Result in data loss, monetary loss and more.
- IBM (2020) found that the average cost of a data breach is \$3.68 million.



SOLUTION

- Password managers eliminate the difficulties that are associated with having unique and complex passwords.
- Password managers store all the passwords in a secured vault that can be accessed with a “master” password.
- No need to remember multiple passwords.
- Greatly reduces the chances of having multiple accounts compromised.

RESEARCH

Most Popular Password Managers:

- KeePass
- LastPass
- 1Password
- Dashlane
- NordPass

Industry Standards:

- Password Encryption
- Hashing used for authentication
- Automatically copy password to clipboard
- Sanitize clipboard
- Generates strong passwords

XECURE PASSWORD MANAGER

Xecure is a
password manager
application for
Windows
computers.



Uses MYSQL
database to store
user data.



Uses encryption and
hashing to protect
user data.

Securely stores and
manages log in
information.



IMPLEMENTATION LANGUAGES & OTHER REQUIREMENTS

System Requirements:

- Windows 10 or Windows 11
- Python 3.9.7
- MySQL Server (8.0.26)

Software:

- MySQL Workbench 8

Libraries & Packages:

- Hashlib
- Cryptography
- Msvcrt
- Mysql.connector
- Random
- Smtplib
- Socket
- Time
- Pyperclip
- Threading
- Getpass

More information and installation instruction on: <https://github.com/Xcar17/Xecure-Password-Manager>

MAJOR FEATURES

- Uses SHA-256 to authenticate users.
- No need to store “master” password.
- Uses AES-256 to encrypt data before sending it to the database.
- Uses prepared statements to prevent SQL injections.



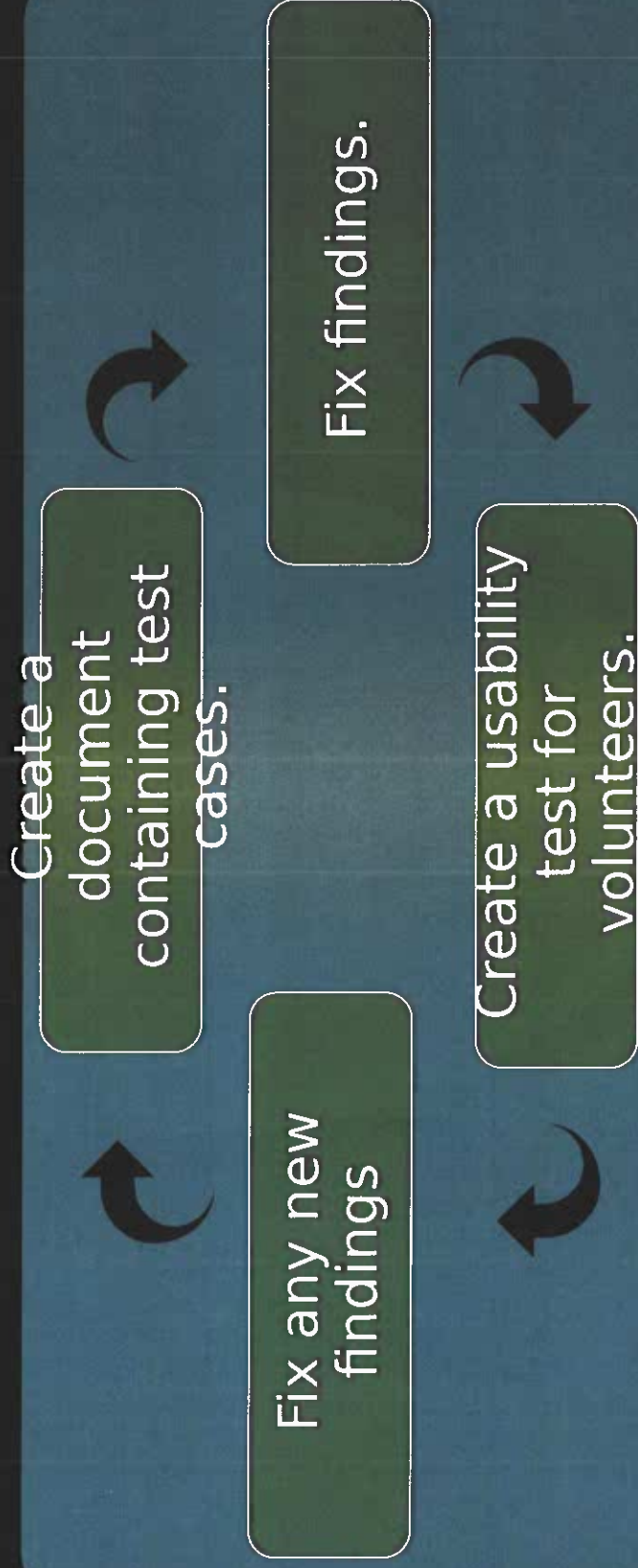
USER FEATURES



- Create unlimited accounts.
- Create and save records.
- Generate strong password for records.
- Retrieve records.
- Automatically copy password to clipboard.
- Update records.
- Account recovery.

[Live Demo...](#)

TEST PLAN



USABILITY TEST RESULTS



- Users wanted a way to back out of menus.
- Users did not want to remember a user ID.
- Verification Codes were too complicated.
- Email addresses were case sensitive.
- Recovery emails needed more details.

TEST CASE RESULTS

- Application does not sanitize clipboard if user force closes the application.
- No email validation when registering.

VULNERABILITIES

- Password not being sanitized after force closing application.
- Unauthorized user gaining access to the system.
- System becoming infected with malware or spyware.
- Database deletion or corruption.
- Shoulder surfing.

FUTURE ENHANCEMENTS

- Email Validation
- Password Strength Detector
- Better Input Validation
- Better Password Rules
- Clipboard Exit/Logout Fix
- Additional Authentication Methods
- Password Verification Prompt
- Better Lockout Feature
- Full Implementation of Password Generator
- Prevent Leading Zeros
- Prevent Force Closing App (Tentative)
- GUI Interface (Tentative)

CHALLENGES & MOTIVATION

Challenges:

- Time
- Encryption/decryption
- Database connectivity
- Python

Motivation:

- Learning about encryption
- Learning python
- Getting comfortable with a database
- Preventing SQL injections
- Increase awareness of password security
- The need for a free and secure Password Manager

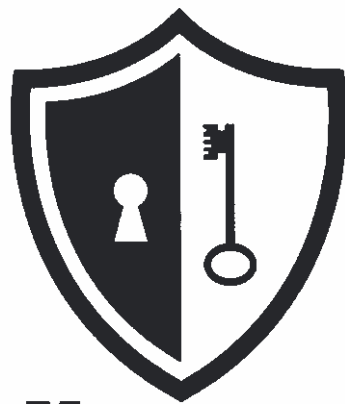


GITHUB LINK

<https://github.com/Xcar17/Xecure-Password-Manager>

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Xecure

